

3D GEOMETRY

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1. LINES

1.1 Vector Equation of a Line

A. Vector Form

There are many shapes in 3D Geometry. We look at one of the simplest (lines), first in Vector form, and then Cartesian Form.

1.1: Point-Vector Form

Given a point P with position vector $\vec{a}$ whose terminal point lies on a line, and a vector $\vec{b}$ parallel to the line, the vector form of the equation of the line is:

$$\vec{r} = \underbrace{\vec{a}}_{\text{Point}} + \underbrace{\lambda}_{\text{Parameter}} \underbrace{\vec{b}}_{\text{Direction Vector}}$$

Consider an arbitrary point Q on the line with position vector $\vec{r}$.

Since $\vec{b}$ is parallel to the line:

$$\vec{PQ} \parallel \vec{b}$$

The two vectors above have the same direction. Hence, one must be a scalar multiple of the other:

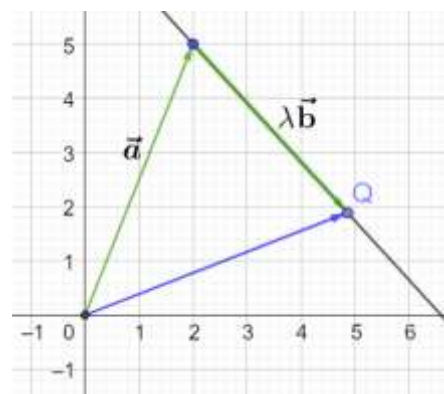
$$\vec{PQ} = \lambda \vec{b}$$

Write $\vec{PQ}$ as a displacement vector:

$$\vec{PQ} = \vec{r} - \vec{a} = \lambda \vec{b}$$

Solve for $\vec{r}$:

$$\vec{r} = \vec{a} + \lambda \vec{b}$$



1.2: The parameter λ

λ is a parameter that can take different values depending on the length of the parallel vector.

- As we will see later, on the applications of the vector equation is to take time as a parameter. For example, in moving objects (or Kinematics).

Example 1.3

- Find the vector equation of a line which passes through the point $(3,4,5)$ and is parallel to the vector $2\hat{i} + 2\hat{j} - 3\hat{k}$. (CBSE 2019)
- Write the equation of the straight line through the point (α, β, γ) and parallel to the z -axis. (CBSE 2014)

Part A

Substitute $\vec{a} = (3,4,5)$, $\vec{b} = (2,2,-3)$ in $\vec{r} = \vec{a} + \lambda \vec{b}$:

$$\vec{r} = (3,4,5) + \lambda(2,2,-3)$$

Part B

Substitute $\vec{a} = (\alpha, \beta, \gamma)$, $\vec{b} = (0,0,1)$ in $\vec{r} = \vec{a} + \lambda \vec{b}$:

$$\vec{r} = (\alpha, \beta, \gamma) + \lambda(0,0,1)$$

1.4: Two Point Form

If $\vec{c}$ and $\vec{a}$ are position vectors of points on the line, the two-point form of the vector equation of a line is:
$$\vec{r} = \vec{a} + \lambda(\vec{c} - \vec{a})$$

Example 1.5

The points $A(4,5,10)$, $B(2,3,4)$ and $C(1,2,-1)$ are three vertices of parallelogram $ABCD$. Find the vector equations of sides AB and BC and find coordinates of point D . (CBSE 2010)

Let O be the origin.

Substitute in $\vec{r} = \vec{a} + \lambda(\vec{b}_1 - \vec{a})$:

$$\vec{AB} = (4,5,10) + \lambda[(2,3,4) - (4,5,10)] = (4,5,10) + \lambda(-2, -2, -6)$$

$$\vec{BC} = (2,3,4) + \lambda[(1,2,-1) - (2,3,4)] = (2,3,4) + \lambda(-1, -1, -5)$$

1.6: Non-uniqueness of the Vector Form of the Equation of a Line

The vector form of the equation of a line is not unique.

Example 1.7

Show from the previous example that

$$\vec{AB} = (4,5,10) + \lambda_1(1,1,3)$$

$$\vec{BC} = (2,3,4) + \lambda_1(1,1,5)$$

$$\vec{AB} = (4,5,10) + \lambda(-2, -2, -6) = (4,5,10) - 2\lambda(1,1,3) = (4,5,10) + \lambda_1(1,1,3)$$

$$\vec{BC} = (2,3,4) - \lambda(1,1,5) = (2,3,4) + \lambda_1(1,1,5)$$

Example 1.8

Find the equation of:

- the z axis.
- the y axis.
- the x axis.
- a line parallel to the z axis, and passing through the point $(1,2,3)$.

Part A

The z axis includes the origin. Hence, $(0,0,0)$ is a point on the line.

The unit vector $\hat{k}$ is parallel to the z axis.

Hence, the equation of the z axis is:

$$\vec{r} = (0,0,0) + \lambda(0,0,1) = \lambda\hat{k}$$

Parts B and C

The equations of the other axes are:

$$\vec{r} = (0,0,0) + \lambda(0,1,0) = \lambda\hat{j}$$

$$\vec{r} = (0,0,0) + \lambda(1,0,0) = \lambda\hat{i}$$

Part B

$(1,2,0)$ is a point on the line. The unit vector $\hat{k}$ is parallel to the z axis. The equation of the line is:

$$\vec{r} = (1,2,0) + \lambda(0,0,1) = \lambda\hat{k}$$

In the above example, the equations that have been found are:

$$\vec{r} = \lambda\hat{k}, \quad \vec{r} = \lambda\hat{j}, \quad \vec{r} = \lambda\hat{i}$$

Is it a good idea to use the same variable ($\vec{r}$) for all three equations, or should we have written something like

$$\vec{a} = \lambda \hat{k}, \quad \vec{b} = \lambda \hat{j}, \quad \vec{c} = \lambda \hat{i}$$

When we write equations in 2D in Cartesian form, we do not use different variables. For example, two lines could be:

$$y = 2x + 3$$
$$y = \frac{1}{2}x + 4$$

And both of the equations use the variables x and y . If we wanted to find the intersection of these lines, we would note that the LHS of each equation is the same, and hence the RHS must also be the same.

$$2x + 3 = \frac{1}{2}x + 4$$

Find the point of intersection of the axes by solving the system of equations:

$$\vec{r} = \lambda \hat{k}, \quad \vec{r} = \lambda \hat{j}, \quad \vec{r} = \lambda \hat{i}$$

We know that the point of intersection is the origin (0,0,0). We prove it by solving the system of equations given.

Note that the LHS of each equation is the same, and hence the RHS must also be the same.

$$\lambda \hat{k} = \lambda \hat{j} = \lambda \hat{i}$$

$\hat{k}$ is purely in the z direction, and no scalar multiple of $\hat{k}$ will change its direction to that of $\hat{j}$ (purely in the y direction) or $\hat{i}$ (purely in the x direction).

Hence, the only possible value of λ is a multiple that has no direction, which means:

$$\lambda = 0 \Rightarrow \text{Point of Intersection} = (0,0,0)$$

Example 1.9: Finding a Point

Find the point with x coordinate 7 that lies on the line $\vec{r} = (2,3,4) + \lambda(1,1,5)$.

Carry out the scalar multiplication and add the vectors on the RHS of the given equation:

$$\vec{r} = (2 + \lambda, 3 + \lambda, 4 + 5\lambda)$$

The point that we want has coordinates (7, y , z). Equate the x coordinates:

$$2 + \lambda = 7 \Rightarrow \lambda = 5$$

Substitute the value of the parameter $\lambda = 5$ to find the value of the y and the z coordinates:

$$y = 3 + \lambda = 3 + 5 = 8$$
$$z = 4 + 5\lambda = 4 + 5(5) = 29$$

Example 1.10: Checking for a Point

$$\vec{r} = (0,2,3) + \lambda(2,1,-5)$$

- A. Explain why the point (1,2,3) does not lie on the above line.
- B. Find a point that does lie on the above line by changing only one of the x , y and z coordinates.

Part A

Carry out the scalar multiplication and add the vectors on the RHS of the given equation:

$$\vec{r} = (2\lambda, 2 + 2\lambda, 3 - 5\lambda)$$

The point that we are investigating is (1,2,3). Equate the x , y and z coordinates:

$$\begin{aligned} 2\lambda &= 1 \Rightarrow \lambda = \frac{1}{2} \\ 2 + 2\lambda &= 2 \Rightarrow \lambda = 0 \\ 3 - 5\lambda &= 3 \Rightarrow \lambda = 0 \end{aligned}$$

Since the value of the parameter is not the same from all three equations, the point does not lie on the line.

Part B

From the y and z coordinates, we get $\lambda = 0$. Find the value of the x coordinate, when $\lambda = 0$:

$$x = 2\lambda = 2(0) = 0$$

The point that we want is:

$$(0, 2, 3)$$

B. Direction Ratios

1.11: Direction Ratios (DR)

If I have the position vectors of two points on a line, then the equation of a line written in standard vector form is

$$\vec{r} = \vec{a} + \lambda(\vec{c} - \vec{a})$$

its direction ratios are

$$\vec{c} - \vec{a} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

The direction ratios, as the name says, indicate the ratio of the change in different coordinates:

$$a : b : c = \Delta x : \Delta y : \Delta z = (x_2 - x_1) : (y_2 - y_1) : (z_2 - z_1)$$

Example 1.12

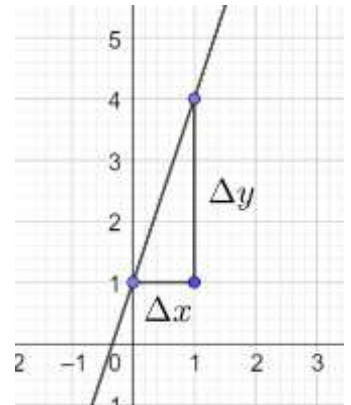
Compare direction ratios with slope for a 2D line.

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \Delta y : \Delta x$$

Slope tells you the rate of change in the y coordinate with respect to rate of change of the x coordinate.

Direction ratios are like slope, except that they compare x, y and z coordinates:

$$\Delta x : \Delta y : \Delta z$$



Example 1.13

Find the direction ratios of the line passing through the points $(1, 2, 3)$ and $(-3, -2, -1)$.

$$\Delta x : \Delta y : \Delta z = (-3 - 1, -2 - 2, -1 - 3) = (-4, -4, -4) = -4(1, 1, 1)$$

Example 1.14

$$\vec{r} = (4, 5, 10) + \lambda(1, 2, 5)$$

- What are the direction ratios of the line above.
- Interpret these direction ratios in terms of change in x coordinate, change in y coordinate, and change in z coordinate.
- If the x coordinate increases by 10, then find the change in the y and the z coordinates.
- Suppose the point $P = (x, y, z)$ lies on the line above. Find the point with x coordinate 2 units greater than P that lies on the line.

Part A

The direction ratios are:

$$1: 2: 5$$

Part B

$$\Delta x: \Delta y: \Delta z = 1: 2: 5$$

Part C

$$\begin{aligned}\Delta x: \Delta y: \Delta z &= 1: 2: 5 = 10: 20: 50 \\ \Delta x &= \text{Change in } x \text{ coordinate} = 10 \\ \Delta y &= \text{Change in } y \text{ coordinate} = 20 \\ \Delta z &= \text{Change in } z \text{ coordinate} = 50\end{aligned}$$

Part D

$$\Delta x: \Delta y: \Delta z = 1: 2: 5 = 2: 4: 10$$

The point that we require is:

$$Q = (x + 2, y + 4, z + 10)$$

C. Direction Cosines

1.15: Direction Angles

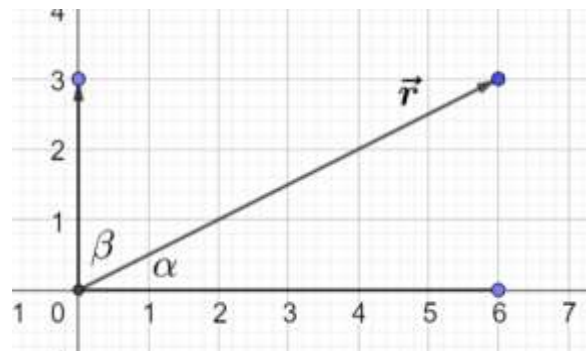
If O is the origin, then the angles made by $\overrightarrow{OP}$ with the positive direction of x -axis, y -axis and z -axis are the direction angles.

Angle with positive direction of	
x - axis	α
y - axis	β
z - axis	γ

1.16: Direction Cosines

The cosines of the direction angles are the direction cosines:

$$\begin{aligned}l &= \frac{x}{|\vec{r}|} = \cos \alpha \\ m &= \frac{y}{|\vec{r}|} = \cos \beta \\ n &= \frac{z}{|\vec{r}|} = \cos \gamma\end{aligned}$$



- Direction cosines are a way of normalizing the direction ratios of a line.
- Direction cosines are always direction ratios, but direction ratios may or may not be direction cosines.

Example 1.17

- If a line makes angles $90^\circ, 135^\circ, 45^\circ$ with the x, y and z axes respectively, find its direction cosines (CBSE 2019)
- Write the direction cosines as a triplet.

Part A

$$\begin{aligned}l &= \cos 90^\circ = 0 \\ m &= \cos 135^\circ = -\frac{1}{\sqrt{2}} \\ n &= \cos 45^\circ = \frac{1}{\sqrt{2}}\end{aligned}$$

Part B

The direction cosines for the line are:

$$(l, m, n) = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

Example 1.18

- Write the direction cosines of a line as a ratio.
- Interpret the ratio.

Part A

$$l : m : n = \cos \alpha : \cos \beta : \cos \gamma = \frac{x}{|\vec{r}|} : \frac{y}{|\vec{r}|} : \frac{z}{|\vec{r}|} = x : y : z$$

Part B

This ratio gives the change in x , y and z for change in the magnitude of the associated vector. In other words, this is a direction ratio.

Example 1.19

Find direction cosines for the axes.

$$\begin{aligned}x - \text{axis} &= (\cos 0, \cos 90^\circ, \cos 90^\circ) = (1, 0, 0) \\y - \text{axis} &= (\cos 90^\circ, \cos 0, \cos 90^\circ) = (0, 1, 0) \\z - \text{axis} &= (\cos 90^\circ, \cos 90^\circ, \cos 0) = (0, 0, 1)\end{aligned}$$

1.20: Sum of Direction Cosines

The sum of the squares of direction cosines for a line is 1.

$$l^2 + m^2 + n^2 = \left(\frac{x}{|\vec{r}|}\right)^2 + \left(\frac{y}{|\vec{r}|}\right)^2 + \left(\frac{z}{|\vec{r}|}\right)^2 = \frac{x^2 + y^2 + z^2}{|\vec{r}|^2} = \frac{x^2 + y^2 + z^2}{x^2 + y^2 + z^2} = 1$$

Example 1.21

- What are the direction cosines of a line that makes equal angles with the coordinate axes? (CBSE 2008, 2009, 2011, 2019)
- If a line makes angles 90° and 60° respectively with the positive directions of X and Y –axes, find the angle which it makes with the positive direction of Z -axis. (CBSE 2017)
- If a line makes angles 90° , 60° and θ with X , Y and Z axes respectively, where θ is an acute angle, then find θ . (CBSE 2015)
- If a line makes angles α , β and γ with the positive direction of coordinate axes, then write the value of $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$. (CBSE 2015C)

Part A

$$l^2 + m^2 + n^2 = 1$$

But the angles are equal, which means their cosines are equal:

$$l = m = n$$

Hence:

$$3l^2 = 1 \Rightarrow l = \pm \frac{1}{\sqrt{3}}$$

The direction cosines are:

$$\left\{\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right\} \text{ OR } \left\{-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right\}$$

Part B

$$\cos^2 90^\circ + \cos^2 60^\circ + \cos^2 \gamma = 1$$

$$0^2 + \left(\frac{1}{2}\right)^2 + \cos^2 \gamma = 1$$

$$\cos^2 \gamma = \frac{3}{4}$$

$$\cos \gamma = \pm \frac{\sqrt{3}}{2}$$

$$\gamma = \{30^\circ, 150^\circ\}$$

Part C

Same as Part B, above except that we reject the angle in the second quadrant.

Part D

$$\begin{aligned} \theta &= 30^\circ \\ \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma &= (1 - \cos^2 \alpha) + (1 - \cos^2 \beta) + (1 - \cos^2 \gamma) \\ &= 3 - \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

Example 1.22

If a unit vector $\hat{a}$ makes angles $\frac{\pi}{3}$ with $\hat{i}$, $\frac{\pi}{4}$ with $\hat{j}$ and $\theta \in \{0, \pi\}$ with $\hat{k}$, then the possible values of θ are: (JEE Main 2019, 9 April, Shift-II)

$$\begin{aligned} l^2 + m^2 + n^2 &= 1 \\ \cos^2\left(\frac{\pi}{3}\right) + \cos^2\left(\frac{\pi}{4}\right) + \cos^2(\theta) &= 1 \\ \frac{1}{4} + \frac{1}{2} + \cos^2(\theta) &= 1 \\ \cos^2(\theta) &= 1 - \frac{3}{4} = \frac{1}{4} \\ \cos \theta &= \pm \frac{1}{2} \\ \theta &= \left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\} \end{aligned}$$

1.23: Converting from Direction Ratios to Direction Cosines

$$\text{Direction Cosines} = l : m : n \Leftrightarrow \text{Direction Ratios} = lX : mX : nX$$

Where X is some constant

$$\left(\frac{\Delta x}{X}\right)^2 + \left(\frac{\Delta y}{X}\right)^2 + \left(\frac{\Delta z}{X}\right)^2 = 1, X = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

$$\frac{\Delta x}{X}, \frac{\Delta y}{X}, \frac{\Delta z}{X}, X = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

Example 1.24

Determine the direction cosines of the line

- A. $\vec{r} = (-2, 3, 7) + \lambda(1, 2, 5)$
- B. $\vec{r} = (-2, 3, 7) + \lambda(3, 4, 5)$

Part A

$$\text{Direction Ratios} = \Delta x : \Delta y : \Delta z = 1 : 2 : 5$$

$$1^2 + 2^2 + 5^2 = 1 + 4 + 25 = 30$$

Divide both sides by 30:

$$\begin{aligned} \frac{1^2 + 2^2 + 5^2}{30} &= \frac{30}{30} = 1 \\ \frac{1^2}{30} + \frac{2^2}{30} + \frac{5^2}{30} &= 1 \\ \left(\frac{1}{\sqrt{30}}\right)^2 + \left(\frac{2}{\sqrt{30}}\right)^2 + \left(\frac{5}{\sqrt{30}}\right)^2 &= 1 \end{aligned}$$

Part B

$$X = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50}$$
$$\text{Direction Cosines} = \left(\frac{3}{\sqrt{50}}, \frac{4}{\sqrt{50}}, \frac{5}{\sqrt{50}} \right)$$

MCQ 1.25

Mark the correct option

Consider the vector equation of a line. Changing the point $\vec{a}$ to another point on the same line will change the direction cosines.

- A. If the vector equation is in point vector form $\vec{r} = \vec{a} + \lambda \vec{b}$
- B. If the vector equation is in two point vector form $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$
- C. Both of the above
- D. None of the above

Changing to another point on the same line keeps the parallel vector the same, and hence the direction ratios (and directions cosines) also the same.

Option D

Example 1.26

Mark all correct options

Two lines are parallel or the same line, if:

- A. Their direction ratios are the same.
- B. Their direction cosines are the same.
- C. The direction ratios of one line are $x:y:z$ and the direction ratios of the second line are $ax:ay:az$.
- D. The direction cosines of one line are a multiple of the direction cosines of the other line.

If two lines have the same direction cosines, they are going in the same direction, and hence they are parallel.

Option B

If two lines have equal direction ratios, then they point in the same direction. Again, the lines are parallel.

Option A

Simplify the direction ratios of both lines, and then the direction ratios will be equal. Again, they point in the same direction, and they are parallel.

Option C

1.2 Cartesian Equation of a Line

A. Cartesian Form

1.27: Cartesian Form

The standard form of the equation of a line is given by:

$$\lambda = \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Substitute the coordinates of the vectors in the point-vector form of the equation of a line:

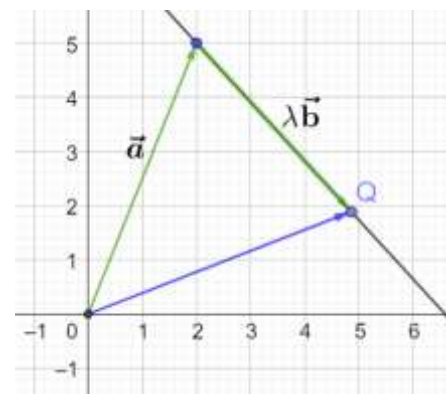
$$\vec{r} = \vec{a} + \lambda \vec{b} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \lambda \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} x_1 + \lambda a \\ y_1 + \lambda b \\ z_1 + \lambda c \end{pmatrix}$$

Equate the components:

$$x = x_1 + \lambda a, \quad y = y_1 + \lambda b, \quad z = z_1 + \lambda c$$

Solve for λ in each equation to get:

$$\lambda = \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$



1.28: Components of the Cartesian Form

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

(a, b, c) are the Direction Ratios of the line

(x_1, y_1, z_1) are the coordinates of a point on the line

Example 1.29

- (Cartesian form) A line passes through the point with position vector $2\hat{i} - \hat{j} + 4\hat{k}$ and is in the direction of the vector $\hat{i} + \hat{j} - 2\hat{k}$. Find the equation of the line in Cartesian form. (CBSE 2019)
- (Cartesian to Vector form) 17: Write the vector equation for the line $\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$. (CBSE 2011)
- (Convert from Vector to Cartesian form)

Part A

Substitute $(x_1, y_1, z_1) = (2, -1, 4)$, $(a, b, c) = (1, 1, -2)$:

$$\frac{x - 2}{1} = \frac{y + 1}{1} = \frac{z - 4}{-2}$$

Part B

Substitute $(x_1, y_1, z_1) = (5, -4, 6)$, $(a, b, c) = (3, 7, 2)$:

$$\vec{r} = \vec{a} + \lambda \vec{b} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 7 \\ 2 \end{pmatrix}$$

1.30: Converting to Standard Form

If the equation of a line is not in standard form, we need to convert it to standard form before proceeding to work with it.

The standard form is

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

- The coefficient of x , y and z is 1.
- The sign of x_1 , y_1 and z_1 is $-ve$.

Example 1.31

A student attempted the question "Write the equation $3x = 4y$ in such a way that the coefficient of both x and y

is 1”;

$$\frac{3x}{3} = \frac{4y}{4} \Rightarrow x = y$$

Is this correct? If it is not correct, explain why and write the corrected version.

We divide the LHS by 3, and the RHS by 4.

But we need to perform equal operations to maintain equality. Hence the above is not correct.

Corrected Version

Divide both sides by $LCM(3,4) = 12$

$$\frac{3x}{12} = \frac{4y}{12} \Rightarrow \frac{x}{4} = \frac{y}{3}$$

Example 1.32

A student attempted the question “Write the equation $3x = 4y = 5z$ in such a way that the coefficient of x , y and z is 1”:

$$\frac{3x}{3} = \frac{4y}{4} = \frac{5z}{5} \Rightarrow x = y = z$$

Is this correct? If it is not correct, explain why and write the corrected version.

Example 1.33

A student attempted the question “Write the equation $3x = 4y$ in such a way that the coefficient of both x and y is 1”:

$$\frac{3x}{3} = \frac{4y}{4} \Rightarrow x = y$$

Is this correct? If it is not correct, explain why and write the corrected version.

Example 1.34

Identify the direction ratios and a point on each line below:

A. $\frac{x-3}{2} = \frac{y+5}{-4} = \frac{2z-1}{6}$
B. $\frac{3x-x_1}{a} = \frac{y_1-2y}{b} = \frac{z_1-z}{c}$

Part A

$$\frac{x-3}{2} = \frac{y+5}{-4} = \frac{\frac{1}{2}(2z-1)}{\frac{1}{2} \times 6} \Rightarrow \frac{x-3}{2} = \frac{y+5}{-4} = \frac{z-\frac{1}{2}}{3} \Rightarrow \underbrace{(2, -4, 3)}_{\text{Direction Ratios}}, \underbrace{\left(3, -5, \frac{1}{2}\right)}_{\text{Point on the Line}}$$

Part B

$$\frac{\frac{1}{3}(3x-x_1)}{\frac{1}{3} \times a} = \frac{\left(-\frac{1}{2}\right)(y_1-2y)}{-\frac{1}{2} \times b} = \frac{-(z_1-z)}{-c} = \frac{x-\frac{x_1}{3}}{\frac{1}{3} \times a} = \frac{y-\frac{y_1}{2}}{-\frac{1}{2} \times b} = \frac{z-z_1}{-c}$$

$$\underbrace{\left(\frac{a}{3}, -\frac{b}{2}, -c\right)}_{\text{Direction Ratios}}, \underbrace{\left(\frac{x_1}{3}, \frac{y_1}{2}, z_1\right)}_{\text{Point on the Line}}$$

Example 1.35

$$\frac{4-x}{2} = \frac{y}{6} = \frac{1-z}{3}$$

- Identify the direction ratios in the above line.
- Use the direction ratios to find a vector parallel to the line.
- Use the direction ratios to find all vectors parallel to the line.

Direction Ratios are 2,6,3
Vector parallel to the line = (2,6,3)
All vectors parallel to the line = $\lambda(2,6,3), \lambda \neq 0$

Is $\lambda \neq 0$ important in the above

$$\text{If } \lambda = 0 \Rightarrow \lambda(2,6,3) = 0(2,6,3) = (0,0,0) \Rightarrow \text{Zero vector}$$

Since the zero vector has no direction, it cannot be parallel to a line.

Example 1.36: Converting to Vector Form

If the Cartesian equation of a line is $\frac{3-x}{5} = \frac{y+4}{7} = \frac{2z-6}{4}$, then write the vector equation for the line. (CBSE 2014)

$$\frac{x-3}{-5} = \frac{y+4}{7} = \frac{z-3}{2}$$

$$\vec{r} = (3, -4, 3) + \lambda(-5, 7, 2)$$

Example 1.37: Finding a point

The point $(3, y_1, z_1)$ lies on the line $\frac{x-2}{3} = \frac{y-3}{1} = \frac{z-1}{2}$. Find $y_1 + z_1$:

- By finding the parameter λ
- By using the direction ratios

Part A

Let $\frac{x-2}{3} = \frac{y-3}{1} = \frac{z-1}{2} = \lambda$, and solve for x, y and z :

$$\frac{x-2}{3} = \lambda \Rightarrow x = 3\lambda + 2,$$

$$\frac{y-3}{1} = \lambda \Rightarrow y = \lambda + 3$$

$$\frac{z-1}{2} = \lambda \Rightarrow z = 2\lambda + 1$$

Substitute $x = 3$, and solve for λ :

$$x = 3\lambda + 2 \Rightarrow 3 = 3\lambda + 2 \Rightarrow \lambda = \frac{1}{3}$$

Solve for y and substitute $\lambda = \frac{1}{3}$:

$$y = \lambda + 3 = \frac{1}{3} + 3 = \frac{10}{3}$$

$$z = 2\lambda + 1 = 2\left(\frac{1}{3}\right) + 1 = \frac{2}{3} + 1 = \frac{5}{3}$$

The point is:

$$\left(3, \frac{10}{3}, \frac{5}{3}\right) \Rightarrow y_1 + z_1 = \frac{10}{3} + \frac{5}{3} = \frac{15}{3} = 5$$

Part B

From the given equation, a point on the line is:

$$(x_2, y_2, z_2) = (2, 3, 1)$$

We wish to find the coordinates of another point on the line:

$$(x_1 = 3, y_1, z_1)$$

$$\Delta x = x_1 - x_2 = 3 - 2 = 1$$

We can find the point on the line without finding the vector form or the parameter λ .

$$DR = \Delta x : \Delta y : \Delta z = 3 : 1 : 2 = 1 : \frac{1}{3} : \frac{2}{3}$$

$$y_1 = y_2 + \Delta y = 3 + \frac{1}{3} = \frac{10}{3}$$

$$z_1 = z_2 + \Delta z = 1 + \frac{2}{3} = \frac{5}{3}$$

Example 1.38: Finding a point

The point $(x_1, -4, z_1)$ lies on the line $\frac{2x-5}{2} = \frac{3-\frac{y}{4}}{3} = \frac{4z-2}{4}$.

- A. By finding the parameter λ
- B. By using the direction ratios

Part B

$$\frac{x - \frac{5}{2}}{1} = \frac{y - 12}{-12} = \frac{z - \frac{1}{2}}{1} \Rightarrow (x_2, y_2, z_2) = \left(\frac{5}{2}, 12, \frac{1}{2}\right)$$

$$\Delta y = y_2 - y_1 = -4 - 12 = -16$$

$$DR = \Delta x : \Delta y : \Delta z = 1 : -12 : 1 = \frac{4}{3} : -16 : \frac{4}{3}$$

$$x_1 = x_2 + \frac{4}{3} = \frac{5}{2} + \frac{4}{3} = \frac{15}{6} + \frac{8}{6} = \frac{23}{6}$$

$$z_1 = z_2 + \frac{4}{3} = \frac{1}{2} + \frac{4}{3} = \frac{3}{6} + \frac{8}{6} = \frac{11}{6}$$

$$x_1 + z_1 = \frac{23}{6} + \frac{11}{6} = \frac{34}{6}$$

Example 1.39

The equation of a line can be written, in point-slope form as $y - y_1 = m(x - x_1)$. Show that it is equivalent to:

$$\frac{y - y_1}{\Delta y} = \frac{x - x_1}{\Delta x}$$

Substitute $Slope = m = \frac{y_2 - y_1}{x_2 - x_1}$:

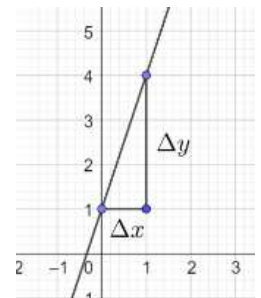
$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1)$$

Rearrange:

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

Substitute $\Delta y = y_2 - y_1$, $\Delta x = x_2 - x_1$:

$$\frac{y - y_1}{\Delta y} = \frac{x - x_1}{\Delta x}$$



1.40: Two Point Form

Given two points (x_1, y_1, z_1) and (x_2, y_2, z_2) that lie on a line, the equation of the line is given by:

$$\lambda = \frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

➤ The two-point Cartesian form in 3D is a generalization of the two-point form in 2D.

$$\lambda = \frac{x - x_1}{\Delta x} = \frac{y - y_1}{\Delta y} = \frac{z - z_1}{\Delta z}$$

Example 1.41

Find the equation of the line passing through points $A(0, 6, -9)$ and $B(-3, -6, 3)$ in Cartesian Form. (CBSE

2010C, Adapted)

Substitute the given data into the two-point form of the equation of a line:

$$\lambda_1 = \frac{x - 0}{-3 - 0} = \frac{y - 6}{-6 - 6} = \frac{z - (-9)}{3 - (-9)}$$

Simplify:

$$\lambda_1 = \frac{x}{-3} = \frac{y - 6}{-12} = \frac{z + 9}{12}$$

Divide throughout by 3, and substitute $\lambda_1 = \lambda$:

$$\lambda = \frac{x - 0}{-1} = \frac{y - 6}{-4} = \frac{z + 9}{4}$$

Example 1.42

Find the equation of the x , y and z axes in Cartesian form.

Equation of z axis

$$\frac{x}{0} = \frac{y}{0} = \frac{z}{1}$$

This does not mean that we are dividing by zero. This is simply a way of communicating that:

$$\Delta x = x_2 - x_1 = 0$$

$$\Delta y = y_2 - y_1 = 0$$

Equation of y axis

$$\frac{x}{0} = \frac{y}{1} = \frac{z}{0}$$

Equation of x axis

$$\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$$

Example 1.43

The vector $(2,6,3)$ is parallel to a line.

- A. If there is a change of 2 units in the x direction, identify the change in the y and z directions, respectively.

$$(\Delta x, \Delta y, \Delta z) = (2, 6, 3)$$

$$\text{Change in the } y \text{ direction} = \Delta y = 6$$

$$\text{Change in the } z \text{ direction} = \Delta z = 3$$

Example 1.44

The equation of the z axis is given by $\frac{x}{0} = \frac{y}{0} = \frac{z}{1}$. Interpret the direction ratios in this equation.

$$(\Delta x, \Delta y, \Delta z) = (0, 0, 1)$$

There is no change in x , (or in y) for any amount, or any change in z .

This makes sense, since no matter the z coordinate, the x and y coordinates of the z axis are fixed to be the origin $= (0,0)$.

B. Direction Cosines

1.45: Converting from Direction Ratios to Direction Cosines

$$\frac{1}{\lambda}(\Delta x, \Delta y, \Delta z), \quad \lambda = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

Direction ratios and direction cosines are both vectors in the same direction.

However, directions cosines have magnitude 1, and directions ratios can have any magnitude.

Hence, dividing the direction ratios by the magnitude scales the direction ratio vector to make it a unit vector.

Example 1.46

The equation of a line is $5x - 3 = 15y + 7 = 3 - 10z$. Write the direction cosines of the line. (CBSE 2015)

Convert the given equation into standard form by making the coefficient of each variable 1. For this, we divide throughout by $LCM(5,15,3) = 30$:

$$\frac{5\left(x - \frac{3}{5}\right)}{30} = \frac{15\left(y - \left(-\frac{7}{15}\right)\right)}{30} = \frac{-10\left(z - \frac{3}{10}\right)}{30}$$

Simplify:

$$\frac{x - \frac{3}{5}}{6} = \frac{y - \left(-\frac{7}{15}\right)}{2} = \frac{z - \frac{3}{10}}{-3}$$

Identify the direction ratios:

$$\text{Direction Ratios} = (\Delta x, \Delta y, \Delta z) = (6, 2, -3)$$

If we think of the direction ratios as a vector, it has magnitude:

$$\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} = \sqrt{6^2 + 2^2 + (-3)^2} = \sqrt{36 + 4 + 9} = 7$$

We want a vector in the same direction, but with magnitude 1, and hence, we divide the vector by the magnitude:

$$\frac{(6, 2, -3)}{7} = \frac{1}{7}(6, 2, -3) = \left(\frac{6}{7}, \frac{2}{7}, -\frac{3}{7}\right)$$

Example 1.47

Find the direction cosines of the line $\frac{4-x}{2} = \frac{y}{6} = \frac{1-z}{3}$. (CBSE 2013)

The given equation is not in standard form. Convert it to standard form:

$$\frac{x-4}{-2} = \frac{y}{6} = \frac{z-1}{-3}$$

Identify the direction ratios:

$$\text{Direction Ratios} = (\Delta x, \Delta y, \Delta z) = (-2, 6, -3)$$

If we think of the direction ratios as a vector, it has magnitude:

$$\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} = \sqrt{(-2)^2 + 6^2 + (-3)^2} = \sqrt{4 + 36 + 9} = 7$$

We want a vector in the same direction, but with magnitude 1, and hence, we divide the vector by the magnitude:

$$\frac{(-2, 6, -3)}{7} = \frac{1}{7}(-2, 6, -3) = \left(-\frac{2}{7}, \frac{6}{7}, -\frac{3}{7}\right)$$

Example 1.48

Write the direction cosines of the lines joining the points (1,0,0) and (0,1,1). (CBSE 2011)

$$DR = (0 - 1, 1 - 0, 1 - 0) = (-1, 1, 1)$$

Magnitude of direction ratio vector

$$= \sqrt{(-1)^2 + 1^2 + 1^2} = \sqrt{3}$$

Hence, direction cosines

$$= \frac{1}{\sqrt{3}}(-1, 1, 1) = \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

C. Collinearity

1.49: Collinearity

If n points lie on the same line, then they are collinear.

Example 1.50

What is the number of distinct lines that pass through:

- A. A point
- B. Two points
- C. Three Points

A point: Infinite

Two points: Exactly One

Three points: One or Zero

Example 1.51

- A. Are two points always collinear?
- B. Are three points always collinear?

Two Points: Yes

Three Points: No

1.52: Checking for Three Collinear Points: Distance Formula

Three points A, B, and C are collinear if

$$l(AB) + l(BC) = l(AC)$$

We can check the distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) using the distance formula:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

1.53: Checking for Three Collinear Points: Parallel Vectors

If three points are collinear, the vectors joining any of two of them must be parallel. This can be interpreted in a couple of ways:

- Parallel vectors will have equivalent direction ratios.
- If two vectors are parallel, one vector will be a scalar multiple of the other.

1.54: Checking for Three Collinear Points: Equation of a Line

To check if three points are collinear, find the equation of the line using any two points, and see if the third point satisfies the equation of the line.

We can use either the Vector equation or the Cartesian Equation.

1.55: Checking for Three Collinear Points: Dot Product

Consider three points A, B and C .

If three points are collinear, the angle between the vectors formed by taking the points pair-wise will be

$$\theta = \{0, 180\}$$

$$\begin{aligned}\cos \theta &= \pm 1 \\ \frac{|\vec{AB}| |\vec{BC}| \cos \theta}{|\vec{AB}| |\vec{BC}|} &= \pm 1 \\ \frac{|\vec{AB}| \cdot |\vec{BC}|}{|\vec{AB}| |\vec{BC}|} &= \pm 1\end{aligned}$$

Take the absolute value on both sides:

$$\left| \frac{|\vec{AB}| \cdot |\vec{BC}|}{|\vec{AB}| |\vec{BC}|} \right| = 1$$

1.56: Checking for Three Collinear Points: Degenerate Triangle

If the area of the triangle formed by the points (x_1, y_1, z_1) , (x_2, y_2, z_2) and (x_3, y_3, z_3) is zero. That is, if:

$$\text{Area of Triangle} = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0$$

1.57: Checking for Three Collinear Points: Cross Product

If two vectors are parallel, their cross product is zero.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

1.3 Multiple Lines

A. Angle between Lines

In the previous section, we looked at the Vector and Cartesian forms of the equation of a line. In this section, we look at some applications related to both forms. There are many formulas: be careful not to mix them up.

1.58: Angle between lines: Definition

The angle between two lines is always acute or right.

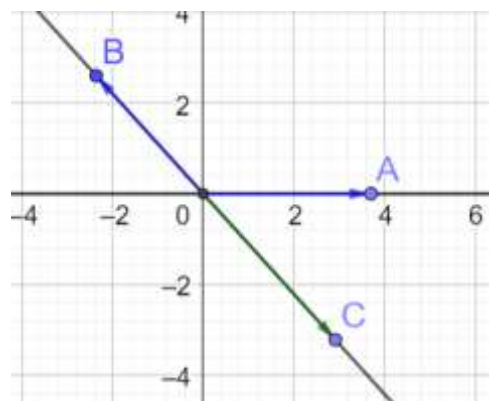
Angle between OA and OC is the angle between the two lines.
(Note that $\angle AOC$ is acute.)

Also, note that $\angle AOC$ and $\angle AOB$ are supplementary. If you know one, you know the other.

1.59: Angle between lines: Formula

Given two lines $\vec{r} = \underbrace{\vec{a}_1 + \lambda \vec{b}_1}_{\text{Line 1}}$ and $\vec{r} = \underbrace{\vec{a}_2 + \mu \vec{b}_2}_{\text{Line 2}}$ with angle θ between them:

$$\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|}$$



$\vec{b}_1$ is a vector parallel to the first line. $\vec{b}_2$ is a vector parallel to the second line. The dot product of the vectors $\vec{b}_1$ and $\vec{b}_2$ is given by:

$$\vec{b}_1 \cdot \vec{b}_2 = |\vec{b}_1| |\vec{b}_2| \cos \theta$$

Rearranging gives us:

$$\cos \theta = \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|}$$

However, note that θ can be acute or obtuse, and if it is obtuse, we wish to find the supplement. In that case, we want:

$$\cos(180 - \theta) = -\frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|}$$

We can combine both the formulas into a single formula by taking the absolute value of the dot product:

$$\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|}$$

Example 1.60

Find the angle between the lines $\vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$ and $\vec{r} = 7\hat{i} - 6\hat{j} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$ (CBSE 2014, 2008)

Use the formula for the angle between two lines:

$$\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|} = \frac{|(3,2,6) \cdot (1,2,2)|}{\sqrt{3^2 + 2^2 + 6^2} \cdot \sqrt{1^2 + 2^2 + 2^2}} = \frac{3 + 4 + 12}{\sqrt{9 + 4 + 36} \cdot \sqrt{1 + 4 + 4}} = \frac{19}{7 \cdot 3} = \frac{19}{21}$$

Take the cos inverse on both sides:

$$\theta = \cos^{-1}\left(\frac{19}{21}\right)$$

Example 1.61

$$\frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1}, \quad \frac{x - x_2}{a_2} = \frac{y - y_2}{b_2} = \frac{z - z_2}{c_2} \Rightarrow \cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Show that given two lines, the angle θ between the two lines above is given by the above formula. Do you need to memorize it?

$$\frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1} = \lambda_1$$

$$x = x_1 + \lambda_1 a_1, \quad y = y_1 + \lambda_1 b_1, \quad z = z_1 + \lambda_1 c_1$$

$$\vec{r} = (x_1, y_1, z_1) + \lambda_1(a_1, b_1, c_1)$$

$$x = x_2 + \lambda_2 a_2, \quad y = y_2 + \lambda_2 b_2, \quad z = z_2 + \lambda_2 c_2$$

$$\vec{r} = (x_2, y_2, z_2) + \lambda_2(a_2, b_2, c_2)$$

Substitute $\vec{b}_1 = (a_1, b_1, c_1)$, $\vec{b}_2 = (a_2, b_2, c_2)$ in the formula for the angle between two lines:

$$\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|} = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Example 1.62

Find the angle between the lines $5x - 3 = 15y + 7 = 3 - 10z$ and $\frac{4-x}{2} = \frac{y}{6} = \frac{1-z}{3}$.

Factor out the leading coefficient for each variable:

$$5\left(x - \frac{3}{5}\right) = 15\left(y + \frac{7}{15}\right) = -10\left(z - \frac{10}{3}\right)$$

Divide throughout by $LCM(5, 15, -10) = 30$:

$$\frac{5\left(x - \frac{3}{5}\right)}{30} = \frac{15\left(y + \frac{7}{15}\right)}{30} = \frac{-10\left(z - \frac{10}{3}\right)}{30}$$

$$\frac{x - \frac{3}{5}}{6} = \frac{y - \left(-\frac{7}{15}\right)}{2} = \frac{z - \frac{10}{3}}{-3}$$

Convert the second line to standard form:

$$\frac{x - 4}{-2} = \frac{y}{6} = \frac{z - 1}{-3}$$

Substitute $\vec{b}_1 = (6, 2, -3)$, $\vec{b}_2 = (-2, 6, -3)$

$$\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|} = \frac{|(6, 2, -3) \cdot (-2, 6, -3)|}{\sqrt{6^2 + 2^2 + (-3)^2} \cdot \sqrt{(-2)^2 + 6^2 + (-3)^2}} = \frac{|-12 + 12 + 9|}{\sqrt{36 + 4 + 9} \cdot \sqrt{4 + 36 + 9}} = \frac{9}{49}$$

$$\theta = \cos^{-1}\left(\frac{9}{49}\right)$$

B. Parallel Lines

1.63: Parallel Lines: Vector Form

Two lines $\vec{r} = \vec{a}_1 + \lambda_1 \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda_2 \vec{b}_2$ are parallel if and only if:

$$\vec{b}_1 = \mu \vec{b}_2$$

If two lines are parallel, then they have the same direction ratios.

In other words, a vector parallel to the first line is a scalar multiple of a vector parallel to the other line.

Example 1.64

- Is the line $\vec{r} = \left(3, -4, -\frac{5}{3}\right) + \lambda_1(2, -1, 3)$ parallel to the line $\vec{r} = (2, -3, -9) + \lambda_2(4, -2, 6)$.
- If the line $\vec{r} = \left(3, -4, -\frac{5}{3}\right) + \lambda_1\left(a, b, \frac{231}{439}\right)$ is parallel to the line $\vec{r} = (2, -3, -9) + \lambda_2\left(4, -2, \frac{521}{1001}\right)$, then find $a: b$.
- Write the vector equation of a line passing through point $(1, -1, 2)$ and parallel to the line whose equation is $\frac{x+3}{3} = \frac{4-y}{5} = \frac{z+8}{6}$. (CBSE 2013)

Part A

$$(2, -1, 3) = \frac{1}{2}(4, -2, 6) \Rightarrow \text{Direction Vectors are } \parallel \Rightarrow \text{Lines are parallel.}$$

Part B

$$(\Delta x, \Delta y, \Delta z) = \left(a, b, \frac{231}{439}\right) = \left(4, -2, \frac{521}{1001}\right)$$

$$a : b = 4 : -2 = 2 : -1$$

Part C

$$\vec{r} = \underbrace{(1, -1, 2)}_{\vec{a_1}} + \lambda_1 \underbrace{(3, 5, 6)}_{\vec{b_1}} = (1, -1, 2) + \lambda(3, 5, 6)$$

1.65: Parallel Lines: Cartesian Form

Two lines $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$ are parallel if their direction ratios are equal. That is if:

$$\underbrace{\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}_{\text{Direction Ratios}} \text{ OR } \underbrace{\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}}_{\text{Direction Cosines}}$$

Example 1.66

Mark the correct Option

Line l_1 has direction ratios $(1, 1, 1)$ and line l_2 has direction ratios $(-1, -1, -1)$.

- A. The lines are parallel since the direction ratios are .
- B. The lines are not parallel since the direction ratios are different.
- C. The lines are parallel since the direction cosines are the same.
- D. The lines are not parallel since the direction cosines are not the same.

$$(-1, -1, -1) = -1(1, 1, 1)$$

The direction ratios are equivalent, and hence the direction cosines will be the same.

Option C

Example 1.67

- A. Are the lines $\frac{x+5}{-3} = \frac{y-2}{4} = \frac{z-4}{-6}$ and $\frac{-10-x}{-3} = \frac{y-1}{-2} = \frac{z-4}{6}$ parallel?
- B. Find the cartesian equation of the line which passes through the point $(-2, 4, -5)$ and is parallel to the line $\frac{x+3}{3} = \frac{4-y}{5} = \frac{z+8}{6}$. (CBSE 2013)

Part A

Write the first line in standard form:

$$\frac{x+5}{-3} = \frac{y-1}{2} = \frac{z-4}{-6} \Rightarrow \text{Parallel Vector} = (-3, 2, -6)$$

Write the second line in standard form:

$$\frac{x+10}{3} = \frac{y-1}{-2} = \frac{z-4}{6} \Rightarrow \text{Parallel Vector} = (3, -2, 6) = -(-3, 2, -6)$$

And since the direction ratios are equivalent, the lines are parallel.

Part B

$$\frac{x+3}{3} = \frac{y-4}{-5} = \frac{z+8}{6} \Rightarrow \parallel \text{ vector} = (3, -5, 6)$$

$$\frac{x+2}{3} = \frac{y-4}{-5} = \frac{z+5}{6}$$

C. Perpendicular Lines

1.68: Perpendicular Lines: Vector Form

Two lines $\vec{r} = \vec{a}_1 + \lambda_1 \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda_2 \vec{b}_2$ are perpendicular *if and only if*:

$$\vec{b}_1 \cdot \vec{b}_2 = 0$$

1.69: Perpendicular Lines: Vector Form

Two lines $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$ are perpendicular *if and only if* the vectors $\vec{b}_1 = (a_1, b_1, c_1)$ and $\vec{b}_2 = (a_2, b_2, c_2)$ are equal. That is:

$$\vec{b}_1 \cdot \vec{b}_2 = 0$$

Example 1.70

Find the value of λ so that the lines $\frac{1-x}{3} = \frac{y-14}{\lambda} = \frac{z-3}{2}$ and $\frac{7-7x}{3\lambda} = \frac{y-5}{1} = \frac{6-z}{5}$ are at right angles. (CBSE 2019)

$$\frac{x-1}{-3} = \frac{y-2}{\frac{\lambda}{7}} = \frac{z-3}{2}, \quad \frac{x-1}{-\frac{3\lambda}{7}} = \frac{y-5}{1} = \frac{z-6}{-5}$$

If the two lines are at right angles, then the dot product of their direction ratios must be zero:

$$(-3)\left(-\frac{3\lambda}{7}\right) + \left(\frac{\lambda}{7}\right)(1) + (2)(-5) = 0 \Rightarrow \frac{9\lambda}{7} + \frac{\lambda}{7} = 10 \Rightarrow \lambda = 7$$

1.71: Two Equations in Three Variables

If we have two equations in three variables

$$x_1a + x_2b + x_3c = 0$$

$$y_1a + y_2b + y_3c = 0$$

the relation between the variables a, b and c is given by:

$$\frac{a}{\begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix}} = \frac{c}{\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}}$$

- We cannot solve for a, b and c since there are two equations and three variables.
- But we can find the relation between the variables.

Example 1.72: Line perpendicular to two lines

Find the vector and cartesian equations of the line through the point $(1, 2, -4)$ and perpendicular to the two lines $\vec{r} = (8, -19, 10) + \lambda(3, -16, 7)$ and $\vec{r} = (15, 29, 5) + \mu(3, 8, -5)$. (CBSE 2015)

A line passing through $(1, 2, -4)$ will have equation:

$$\vec{r} = (1, 2, -4) + \gamma(a, b, c)$$

Since the line is perpendicular to the two given lines, we must have:

$$(a, b, c) \cdot (3, -16, 7) = 0 \Rightarrow 3a - 16b + 7c = 0$$

$$(a, b, c) \cdot (3, 8, -5) = 0 \Rightarrow 3a + 8b - 5c = 0$$

The relation between a, b, c is then:

$$\frac{a}{\begin{vmatrix} -16 & 7 \\ 8 & -5 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 3 & 7 \\ 3 & -5 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 3 & -16 \\ 3 & 8 \end{vmatrix}}$$

$$\frac{a}{(-16)(-5) - (8)(7)} = \frac{-b}{(3)(-5) - (3)7} = \frac{c}{(3)(8) - (-16)(3)}$$

Simplify:

$$\frac{a}{80 - 56} = \frac{b}{15 + 21} = \frac{c}{24 + 48} \Rightarrow \frac{a}{24} = \frac{b}{36} = \frac{c}{72} \Rightarrow \frac{a}{2} = \frac{b}{3} = \frac{c}{6}$$

Let the above equality be equal to the parameter γ :

$$\frac{a}{2} = \frac{b}{3} = \frac{c}{6} = \gamma \Rightarrow a = 2\gamma, b = 3\gamma, c = 6\gamma$$

Finally, the vector equation that we want is:

$$\vec{r} = (1, 2, -4) + \gamma(2, 3, 6)$$

1.73: Line perpendicular to Two Lines

A line perpendicular to two lines is parallel to the cross product of the two lines.

Example 1.74

Find the vector and cartesian equations of the line through the point $(1, 2, -4)$ and perpendicular to the two lines $\vec{r} = (8, -19, 10) + \lambda(3, -16, 7)$ and $\vec{r} = (15, 29, 5) + \lambda(3, 8, -5)$

The line that we want will be parallel to the cross product of the direction ratios of the two given lines. Hence, we want to find:

$$\begin{aligned} \vec{b}_1 \times \vec{b}_2 &= (3, -16, 7) \times (3, 8, -5) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix} \\ &= \begin{vmatrix} -16 & 7 \\ 8 & -5 \end{vmatrix} \hat{i} - \begin{vmatrix} 3 & 7 \\ 3 & -5 \end{vmatrix} \hat{j} + \begin{vmatrix} 3 & -16 \\ 3 & 8 \end{vmatrix} \hat{k} \\ &= [(-16)(-5) - (8)(7)]\hat{i} - [(3)(-5) - (3)7]\hat{j} + [(3)(8) - (-16)(3)]\hat{k} \\ &= (80 - 56)\hat{i} - (15 + 21)\hat{j} + (24 + 48)\hat{k} \\ &= 24\hat{i} + 36\hat{j} + 72\hat{k} \\ &= 12(2\hat{i} + 3\hat{j} + 6\hat{k}) \\ &= 12(2, 3, 6) \end{aligned}$$

The line passing through $(1, 2, -4)$ and with direction vector $12(2, 3, 6)$ will have equation:

$$\vec{r} = (1, 2, -4) + \gamma(2, 3, 6)$$

D. Skew and Intersecting Lines

1.75: Skew Lines

Lines which are neither parallel nor intersecting are skew lines.

The minimum number of dimensions needed for skew lines is 3, since Lines on a 2D coordinate plane are either parallel or intersecting. There is no third possibility.

Example 1.76

Show that the lines $\vec{r} = (1, 1, -1) + \lambda(3, -1, 0)$ and $\vec{r} = (4, 0, -1) + \mu(2, 0, 3)$ intersect. Also, find their point of intersection. (CBSE 2014)

$$\begin{aligned} \vec{r} &= (1 + 3\lambda, 1 - \lambda, -1) \\ \vec{r} &= (4 + 2\mu, 0, -1 + 3\mu) \end{aligned}$$

Equate the y coordinates:

$$1 - \lambda = 0 \Rightarrow \lambda = 1$$

Equate the z coordinates:

$$-1 = -1 + 3\mu \Rightarrow \mu = 0$$

Substitute $\lambda = 1, \mu = 0$ in the x coordinates

$$LHS = 1 + 3\lambda = 4, RHS = 4 + 2\mu = 4 + 2(0) = 4 \Rightarrow \text{Verified}$$

Substitute $\mu = 0$ to find the point of intersection:

$$(4 + 2\mu, 0, -1 + 3\mu) = (4, 0, -1)$$

(Continuation) Example 1.77:

Find the value of λ so that the lines $\frac{1-x}{3} = \frac{7y-14}{\lambda} = \frac{z-3}{2}$ and $\frac{7-7x}{3\lambda} = \frac{y-5}{1} = \frac{6-z}{5}$ are at right angles.

- Using the value of λ that you found check if the lines are intersecting or not. (CBSE 2019)
- Are these lines parallel? What kind of lines are these?

Part A

Substitute the value of $\lambda = 7$ that we found earlier:

$$\frac{x-1}{-3} = \frac{y-2}{1} = \frac{z-3}{2} = t, \quad \frac{x-1}{-3} = \frac{y-5}{1} = \frac{z-6}{-5} = T$$

Writing them in parametric form:

$$(x, y, z) = (1 + 3t, 2 + t, 3 + 2t)$$

$$(x, y, z) = (1 + 3T, 5 + T, 6 - 5T)$$

$$1 + 3t = 1 + 3T \Rightarrow t = T$$

Substituting $t = T$

$$2 + t = 5 + T \Rightarrow 2 = 5 \Rightarrow \text{No Solutions.}$$

Hence, the lines do not intersect.

Part B

$$(-3, 1, 2) \neq (-3, 1, -5) \Rightarrow \text{Lines are not parallel} \Rightarrow \text{Skew Lines}$$

E. Distance between Lines

1.78: Distance between Lines

The shortest distance between the lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ is given by:

$$d = \frac{|(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}_1 \times \vec{b}_2|}$$

Example 1.79

37: Find the shortest distance between the lines whose vector equations are $\vec{r} = (1, 1, 0) + \lambda(1, -3, 2)$ and $\vec{r} = (4, 5, 6) + \mu(2, 3, 1)$. (CBSE 2008, 2014)

1.80: Distance between Parallel Lines

The shortest distance between two parallel lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}$ is given by:

$$d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$$

If two lines are parallel, then they have the same direction ratios.

Example 1.81

40: Find the distance between the lines $\vec{r} = (1, 2, -4) + \lambda(2, 3, 6)$ and $\vec{r} = (3, 3, -5) + \mu(4, 6, 12)$. (CBSE 2014)

F. Distance Formula

1.82: Distance Formula

The distance between $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example 1.83

53: Find the points on the line $\frac{x+2}{3} = \frac{y+1}{2} = \frac{z-3}{2}$ at a distance of 5 units from the point $P(1, 3, 3)$. (CBSE 2010)

We want the distance between

$$Q(3\lambda - 2, 2\lambda - 1, 2\lambda + 3) \text{ and } P(1, 3, 3)$$

Using the distance formula, we get:

$$\sqrt{(3\lambda - 2 - 1)^2 + (2\lambda - 1 - 3)^2 + (2\lambda + 3 - 3)^2} = 5$$

G. Working with Points

1.84: Foot of Perpendicular

1.85: Length of Perpendicular

Example 1.86

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Consider the point $P(3, -1, 11)$ and the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Find the:

- coordinates of foot of perpendicular from P to the line.
- equation of the perpendicular from P to the line.
- length of the perpendicular from P to the line.
- Image of P in the line. (CBSE 2011, Adapted)

2. PLANES

2.1 Point Form

A. Point Form

2.1: Plane

A two-dimensional surface that extends to infinity in all directions is a plane.

- For example, the (x, y) coordinate plane is an example of a plane.

2.2: Normal

In maths, normal is used to indicate perpendicularity.

2.3: Vector Equation of a Plane

If $P_0(x_0, y_0, z_0)$ is a point on a plane, $\vec{n} = (A, B, C)$ is a vector normal to the plane, and $P(x, y, z)$ is a general point on the plane, then the equation of the plane is:

$$\vec{n} \cdot \overrightarrow{P_0P} = 0$$

The normal vector is perpendicular to any vector that lies on the plane.

The dot product of two perpendicular vectors is zero.

Since we already have one point on the plane, the locus of the plane is the set of all points $P(x, y, z)$ that satisfy the equation:

$$\vec{n} \cdot \overrightarrow{P_0P} = 0$$

Concept Check 2.4

Why does the equation of a plane use a vector perpendicular to the plane, and not one parallel to it?

A vector parallel to the plane will not be in a unique direction. Rotating a vector while keeping it in the plane is possible.

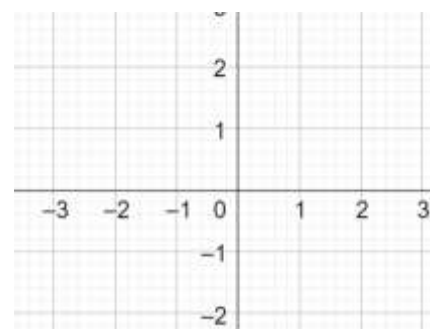
A vector perpendicular to the plane will have unique direction ratios, and hence will be unique.

For example, consider the coordinate plane shown alongside.

x – axis is parallel to the plane

y – axis is parallel to the plane

But the direction ratios for the two are not the same.



Example 2.5

The vector equation of a plane requires a point on the plane, and a vector perpendicular to it. For a given plane, is this vector unique?

Consider the (x, y) coordinate plane.

If we extend this to three dimensions (see diagram), the z –axis is perpendicular to the x and y axes.

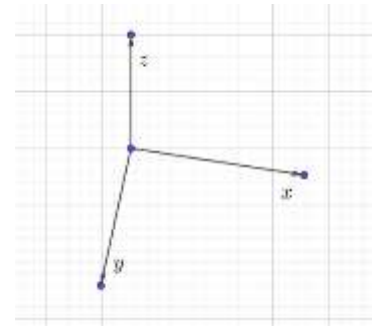
A vector that lies on the z – axis has the form:

$$(0, 0, z), z \in \mathbb{R}, z \neq 0$$

Specifically, the unit vector in the direction of the z –axis is perpendicular to the (x, y) coordinate plane.

$$\hat{k} = (0, 0, 1)$$

And so is any non-zero scalar multiple of $\hat{k}$.



Example 2.6

The vector $\vec{a}$ is perpendicular to a plane. The point B lies on the plane. Determine the equation of the plane.

Let $P(x, y, z)$ be a general point on the plane. Then, the equation of the plane is:

$$\vec{a} \cdot \overrightarrow{BP} = 0$$

2.7: Direction Ratios

The direction ratios of a vector normal to the plane are the components of a vector normal to the plane:

$$DR = (A, B, C) = \vec{n} \text{ in the equation } \vec{n} \cdot \overrightarrow{P_0P} = 0$$

Concept Check 2.8

In the equation of a plane, why do we use direction ratios of a vector perpendicular to the plane, and not those of the plane itself?

If we want a vector on the plane, it will have direction ratios which are not unique.

However, a vector perpendicular to the plane has unique direction ratios.

2.9: Point Form: Cartesian Equation

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Expand the equation $\vec{n} \cdot \overrightarrow{P_0P} = 0$:

$$(A, B, C) \cdot (x - x_0, y - y_0, z - z_0) = 0$$

Multiply using the dot product. This gives us the **Cartesian Equation**:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Concept Check 2.10

Mark all correct options

The equation of a plane has variables whose nature is:

- A. Linear
- B. Quadratic
- C. Trigonometric
- D. Exponential
- E. Logarithmic

Option A: Linear

B. Finding the Equation of a Plane

Example 2.11

Find the equation of a plane passing through the point $(1,3,7)$, and with normal vector $(7,3,1)$.

$$(7,3,1) \cdot (x - 1, y - 3, z - 7) = 0$$

Example 2.12

Answer each part separately

Determine the equation of a plane that meets the conditions below. Is the plane unique.

- A. (x, y) coordinate plane.
- B. (x, z) coordinate plane.
- C. (y, z) coordinate plane.
- D. plane parallel to the (x, y) coordinate plane, and passing through $(1,2,3)$
- E. plane perpendicular to the (x, y) coordinate plane, and passing through $(1,2,3)$

Part A

The origin lies on the (x, y) coordinate plane with coordinates

$$O(0,0,0)$$

The direction ratios of the normal vector to the coordinate plane are:

$$(0,0,1)$$

Hence, the equation of the plane is:

$$(0,0,1) \cdot (x - 0, y - 0, z - 0) = 0$$
$$z = 0$$

Part B

$$y = 0$$

Part C

$$x = 0$$

Part D

$$z = 3$$

Part E

Two equations that work are:

$$x = 1$$

$$y = 2$$

Hence, the plane is not unique.

The planes in parts A-D are unique.

C. Defining a Plane

Example 2.13

What is the number of distinct planes that pass through:

- A. A point
- B. Two points
- C. Three collinear points
- D. Three non-collinear points

- E. Four points, three of which are collinear
- F. Four points, *exactly* three of which are collinear
- G. Four points, no three of which are collinear
- H. Four points
- I. A line and a point
- J. A line and a point not on the line
- K. Two coplanar lines
- L. Two skew lines

Part A

Through one point, we have an infinite number of planes.

A point: Infinite

Part B

Two points define a line, and an infinite number of planes will pass through a line.

Part C

Two collinear points define a line, and an infinite number of planes will pass through a line.

Part D

Three non-collinear points define a plane, and hence exactly one plane will pass through.

Part E

Three collinear points will define a line.

If 4th point is collinear with the other three

Infinite

If 4th point is non-collinear with the other three

One

Part F

This is the second case of Part E.

One

Part G

Zero or One

Part H

Zero or One

Part I

Infinite or One

Part J

One

Part K

One or Infinite

Part L

Zero

Example 2.14

Mark the correct option

Three points define a:

- A. Circle
- B. Triangle
- C. Line
- D. Plane
- E. Either $\{A, B, D\}$ or $\{C\}$

Note:

- Exactly one triangle has its vertices as three non-collinear points and hence three non collinear points define a triangle.
- Exactly one circle passes through three non-collinear points and hence three non collinear points define a circle.

In general, three non-collinear points define a plane, a triangle and the circle that passes the vertices of the triangle.

Example 2.15

What is the number of planes that pass through the points:

- A.

2.16: Defining a Plane

- Through two points (which define a line), infinite number of planes will pass.
- Through three or more collinear points (which define a line), infinite number of planes will pass.
- Through three non-collinear points, exactly one plane will pass. Hence, three non-collinear points define a plane.

2.17: Four Points

- If you have four non-collinear points, any three of them will define a plane.
- The fourth point may or may not lie on the plane determined by the first three points.
- Hence, in general four points may or may not be coplanar.

Example 2.18

Find a nonzero vector that is perpendicular to both of the vectors $(2,3,2)$ and $(4,9,5)$. (Phillips Exeter, Math-3, 2021/537)

Let the vector we want be (x, y, z) . Dot product of two perpendicular vectors is zero:

$$(2,3,2)(x, y, z) = 0 \Rightarrow 2x + 3y + 2z = 0$$

$$(4,9,5)(x, y, z) = 0 \Rightarrow 4x + 9y + 5z = 0$$

Note that we have three variables and two equations. Since, the number of equations is less than the number of variables, the system of equations is *underdetermined*, and it may have infinite solutions.

Since the system is underdetermined, we substitute $z = 1$, and solve the resulting system of two equations in two variables:

$$2x + 3y = -2, \quad 4x + 9y = -5$$

Multiply Equation I by 2, and subtract the new equation from Equation II:

$$0 + 3y = -1 \Rightarrow y = -\frac{1}{3}$$

Substitute $y = -\frac{1}{3}$:

$$4x + 6\left(-\frac{1}{3}\right) = -4 \Rightarrow 4x - 2 = -4 \Rightarrow 4x = -2 \Rightarrow x = -\frac{1}{2}$$

Hence, if $z = 1$, we have determined unique values of x and y . Hence, one vector that meets the conditions is:

$$\left(-\frac{1}{2}, -\frac{1}{3}, 1\right)$$

If we wish to eliminate fractions, we can multiply by 6 to get:

$$= 6\left(-\frac{1}{2}, -\frac{1}{3}, 1\right) = (-3, -2, 6)$$

D. Parallel Planes

2.19: Parallel Planes

Two planes are parallel if they are oriented in the same direction.

This definition should make geometric sense.

Concept Check 2.20

Mark all correct options

Two planes must be parallel if:

- A. the normal vectors have equal direction cosines
- B. the normal vectors have equivalent direction ratios
- C. the normal vectors have dot product zero
- D. the normal vectors have dot product one
- E. the normal vectors have cross product zero
- F. the normal vectors have a unit vector as the cross product
- G. the normal vector of one plane is a scalar multiple of the other

Two planes are parallel when they are oriented in the same direction. We can check this in a few ways.

Normal vectors of the two planes have the same direction cosines. Direction cosines are precise. Hence, they must be the *same*.

Option A is correct

Normal vectors of the two planes have equivalent direction ratios. We can check this in a couple of ways:

- Convert direction ratios to direction cosines and see if they are equal.
Option B is correct
- See if the normal vector of one plane is a non-zero. scalar multiple of the other
Option G is correct
- If the cross product is zero, then the vectors are parallel.
Option E is correct

If the dot product is zero, the vectors are perpendicular. If the dot product is 1, the vectors may or not be parallel.

Options C and D are not correct

Hence, the final answer is:

Options A, B, E, G

2.21: Parallel Planes: Working Rule

Two planes are parallel if they are oriented in the same direction. So they are parallel if any of the following holds:

- Normal vectors of the two planes have the same direction cosines.
- Normal vectors of the two planes have equivalent direction ratios.
- Cross product of the normal vectors is zero.
- Normal vector of one plane is a scalar multiple of the other

Example 2.22

Plane P is parallel to $-4x + 5y - 2z = 37$ and passes through $(-2, 8, 5)$. Write the Vector and Cartesian equations of P in point form.

Vector Form

The Vector equation of a plane in point form is:

$$\vec{n} \cdot \overrightarrow{P_0P} = 0$$

Equation I

Note that $\vec{n} = (-4, 5, -2)$ and use $P_0(x_0, y_0, z_0) = (-2, 8, 5)$ and $P(x, y, z)$ to find:

Equation II

$$\underline{\overrightarrow{P_0P} = (x + 2, y - 8, z - 5)}$$

Equation III

Substitute Equation II and III in Equation I to get the Vector equation in point form:

$$(-4, 5, -2) \cdot (x + 2, y - 8, z - 5) = 0$$

Cartesian Form

To convert to the Cartesian form, expand the LHS using the dot product definition:

$$-4(x + 2) + 5(y - 8) - 2(z - 5) = 0$$

2.23: Parallel Planes: Converse

If two planes are parallel:

- Normal Vectors have equivalent direction ratios
- Normal Vectors have equal direction cosines
- Cross product of normal vectors is zero
- Normal vector of one plane is a scalar multiple of the other

Example 2.24

The plane P passes through the point (a, b, c) and is parallel to the plane $\vec{r} \cdot (1, 1, 1) = 2$. Write the Vector and Cartesian equations of P in point form. (CBSE 2014, Adapted)

We have that plane $\vec{r} \cdot (1, 1, 1) = 2$ is parallel to plane P . Since the two planes are parallel, the direction ratios of their normal vectors are equivalent. Hence, the direction ratios of the normal vector to plane P are:

$$\vec{n} = (1, 1, 1)$$

Vector Form

The Vector equation of a plane in point form is:

$$\underline{\vec{n} \cdot \overrightarrow{P_0P} = 0}$$

Equation I

Note that $\underline{\vec{n} = (1, 1, 1)}$ and use $P_0(x_0, y_0, z_0) = (a, b, c)$ and $P(x, y, z)$ to find:

$$\underline{\overrightarrow{P_0P} = (x - a, y - b, z - c)}$$

Equation III

Substitute Equation II and III in Equation I to get the Vector equation in point form:

$$(1, 1, 1) \cdot (x - a, y - b, z - c) = 0$$

Cartesian Form

To convert to Cartesian form, expand the LHS using the dot product definition:

$$(x - a) + (y - b) + (z - c) = 0$$

2.2 General Form

A. General Form

2.25: General Form: Cartesian Equation:

$$Ax + By + Cz = D$$

Begin with the Cartesian Equation in Point Form:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Open the parentheses

$$Ax - Ax_0 + By - By_0 + Cz - Cz_0 = 0$$

Move the constant terms to the RHS to get:

$$Ax + By + Cz = Ax_0 + By_0 + Cz_0$$

Use a change of variables. Substitute $D = Ax_0 + By_0 + Cz_0$ to get:

$$Ax + By + Cz = D$$

Example 2.26

Write a Cartesian equation in general form for the plane that contains the point $A = (4, 5, -3)$ and that is perpendicular to the line through $B = (5, -2, -2)$ and $C = (7, 1, 4)$. (Phillips Exeter, Math-3, 2021/548)

The normal vector is:

$$\overrightarrow{CB} = (5 - 7, -2 - 1, -2 - 4) = (-2, -3, -6) = -(2, 3, 6)$$

Let $P(x, y, z)$ be a general point on the plane.

Dot product of normal vector and a vector on the plane is zero:

$$\begin{array}{ccc} \overrightarrow{CB} & \cdot & \overrightarrow{AP} = 0 \\ \text{Normal Vector} & & \text{Vector on the Plane} \end{array}$$
$$-(2, 3, 6) \cdot (x - 4, y - 5, z + 3) = 0$$

Use the dot product:

$$(2)(x - 4) + (3)(y - 5) + (6)(z + 3) = 0$$

Expand:

$$2x - 8 + 3y - 5 + 6z + 18 = 0$$

Simplify:

$$2x + 3y + 6z = 5$$

Example 2.27

Find the value of λ such that the line $\frac{x-2}{6} = \frac{y-1}{\lambda} = \frac{z+5}{-4}$ is perpendicular to the plane $3x - y - 2z = 7$. (CBSE 2010)

The normal vector to the plane $3x - y - 2z = 7$ is

$$(3, -1, -2)$$

The direction ratios of the given line are:

$$(6, \lambda, -4)$$

Since the line is perpendicular to the plane, its direction ratios are proportional to the normal vector:

$$2(3, -1, -2) = (6, -2, -4) = (6, \lambda, -4)$$

$$\lambda = -2$$

2.28: General Form: Vector Equation

If $\vec{n}$ is a vector normal to the plane, $\vec{r}$ is a point on the plane, and D is a scalar, then the general form of the vector equation of a plane is:

$$\vec{n} \cdot \vec{r} = D$$

Begin with the Cartesian Equation of a plane in general form:

$$Ax + By + Cz = D$$

Convert this into vector form as well:

$$(A, B, C) \cdot (x, y, z) = D$$

Substitute $\vec{n} = (A, B, C)$, $\vec{r} = (x, y, z)$:

$$\vec{n} \cdot \vec{r} = D$$

Example 2.29

$$(1, 2, -3) \cdot (x - 3, y - 7, z + 4) = 0$$

In the equation of the plane above, identify:

- The direction ratios of a vector normal to the plane.
- A point on the plane

$$\begin{aligned} \text{Direction Ratios} &= (1, 2, -3) \\ \text{Point on the Plane} &= P_0 = (3, 7, -4) \end{aligned}$$

Example 2.30

Write the vector equation of the line passing through $(1, 2, 3)$ and perpendicular to the plane $\vec{r} \cdot (1, 2, -5) + 9 = 0$. (CBSE 2015)

The vector equation of a line is:

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

Substitute $\vec{a} = (1, 2, 3)$, $\vec{b} = (1, 2, -5)$ in

$$\vec{r} = (1, 2, 3) + \lambda(1, 2, -5)$$

Example 2.31

Find the equation of the plane passing through the points $P(8, 3, -2)$, $Q(-2, -5, -1)$ and $R(2, -4, 3)$.

Find a normal vector

Let the normal vector to the plane be $\vec{n} = (a, b, c)$. Then $\vec{n}$ must be perpendicular to any vector in the plane. And hence their dot product must be zero:

$$\begin{aligned} (a, b, c) \cdot \vec{PQ} &= (a, b, c) \cdot (-10, -8, 1) = -10a - 8b + c = 0 \\ (a, b, c) \cdot \vec{PR} &= (a, b, c) \cdot (-6, -7, 5) = -6a - 7b + 5c = 0 \end{aligned}$$

There are three equations and two variables. Let $a = 1$ in the above system:

$$\begin{aligned} -10 - 8b + c &= 0 \\ -6 - 7b + 5c &= 0 \end{aligned}$$

Solving the above system gives us the values A, B, C and hence the direction ratios of the normal vector:

$$(A, B, C) = \left[1, -\frac{4}{3}, -\frac{2}{3} \right]$$

Find the Constant

Substitute $P(8, 3, -2)$ and the normal $(A, B, C) = \left[1, -\frac{4}{3}, -\frac{2}{3} \right]$ into the equation of a plane to find D :

$$Ax + By + Cz = D \Rightarrow 1(8) + \left(-\frac{4}{3}\right)(3) - \left(\frac{2}{3}\right)(-2) = \frac{16}{3} = D$$

Find the Equation of the Plane

Substitute $(A, B, C) = \left[1, -\frac{4}{3}, -\frac{2}{3} \right]$ and $D = \frac{16}{3}$ into the equation of a plane:

$$Ax + By + Cz = D \Rightarrow 1x - \frac{4}{3}y - \frac{2}{3}z = \frac{16}{3} \Rightarrow 3x - 4y - 2z = 16$$

- A. In the above solution, after finding the equations $-10a - 8b + c = 0$ and $-6a - 7b + 5c = 0$, we substituted $a = 1$. Why did we choose 1? Can you substitute any other value?
- B. In the system of the equations, if you substitute $a = 0$, what values of b and c do you get? Are they correct? How do you interpret them?

Part A

There are two parts to this question. The first is, why is the substitution valid.

From an algebraic point of view:

There are three variables and two equations. Hence, you will not be able to arrive at a unique solution. Hence, it makes sense to substitute one variable.

From the geometry point of view:

There will be an infinite number of normal vectors, and two equations in three variables is enough to give us the direction ratios, but not a unique vector.

The second part is, why choose $a = 1$ as the substitution.

We choose $a = 1$ to make the calculations simple.

Part B

Try making the substitution $a = 0$:

$$\begin{aligned}-8b + c &= 0 \Rightarrow c = 8b \\ -7b + 5c &= 0 \Rightarrow c = -\frac{7}{5}b \\ 8b &= -\frac{7}{5}b \Rightarrow b = 0 \Rightarrow c = 0\end{aligned}$$

We get

$$b = c = 0$$

We are looking for the normal vector. The normal vector gives us the direction ratios. If $a = 0$, then the change in the x -direction is zero, and hence (in this case), the change in the other two directions is also zero.

The information that we get is correct, but not useful to us for determining the normal vector.

B. Perpendicular Planes

2.32: Perpendicular Planes

Planes are perpendicular when:

- The dot product of their normal vectors is zero

For two vectors to be perpendicular their normal vectors must be perpendicular to each other. Hence, checking their dot product is sufficient.

Example 2.33

Consider plane P . Is a plane perpendicular to P and one point on P sufficient to define P ?

Suppose P is perpendicular to the (x, y) coordinate plane, and it passes through $(0, 0, 0)$. However, if you rotate you will get an infinite number of planes.

Example 2.34

Write the equation for a plane perpendicular to $-4x + 5y - 2z = 37$, and passing through $(-2, 8, 5)$.

Find the normal vector

Let the normal vector be (a, b, c) . Then we know that:

$$\underbrace{(-4, 5, -2) \cdot (a, b, c) = 0}_{\text{Equation I}}$$

Since we have only one equation in three variables, we can substitute any value that we want for the first two, and find the corresponding value of the third.

Specifically, substitute $a=1$, $b=1$ in Equation I:

$$(-4, 5, -2) \cdot (1, 1, c) = 0$$

Expand using the dot product definition and solve for c :

$$-4 + 5 - 2c = 0 \Rightarrow c = \frac{1}{2}$$

Find the equation of the plane

The Cartesian equation of a plane in point form is:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Substitute the normal vector:

$$\begin{aligned} \left(1, 1, \frac{1}{2}\right) \cdot (x + 2, y - 8, z - 5) &= 0 \\ (x + 2) + (y - 8) + (0.5z - 2.5) &= 0 \\ 2x + 2y + z &= 13 \end{aligned}$$

Is the answer you have found a unique plane.

No.

Example 2.35

Write the equation for the plane perpendicular to $-4x + 5y - 2z = 37$, and passing through $(-2, 8, 5)$ and $(1, 1, 1)$.

Use the normal vector to find one equation

Let the normal vector be (a, b, c) . Then we know that:

$$(-4, 5, -2) \cdot (a, b, c) = 0$$

Substitute $a = 1$ in Equation I:

$$(-4, 5, -2) \cdot (1, b, c) = 0$$

Expand using the dot product definition and solve for c :

$$\underbrace{-4 + 5b - 2c = 0}_{\text{Equation I}}$$

Use a vector that lies on the plane to find another equation

Substitute the normal vector in $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

$$(1, b, c) \cdot (-2 - 1, 8 - 1, 5 - 1) = 0$$

$$(1, b, c) \cdot (-3, 7, 4) = 0$$

$$\underbrace{-3 + 7b + 4c = 0}_{\text{Equation II}}$$

Solve the system of equations

Multiply Equation I by 2 to get $\underbrace{-8 + 10b - 4c = 0}_{\text{Equation III}}$ and add Equations II and III:

$$-11 + 15b = 0 \Rightarrow b = \frac{11}{15}$$

Substitute $b = \frac{11}{15}$ in Equation I:

$$-4 + 5\left(\frac{11}{15}\right) - 2c = 0 \Rightarrow -9 - 4c = 0 \Rightarrow c = \frac{9}{4}$$

Find the equation of the plane

Hence, substitute $\vec{n} = (a, b, c) = \left(1, \frac{11}{15}, \frac{9}{4}\right)$, $P_0(1, 1, 1)$ in $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$
 $\left(1, \frac{11}{15}, \frac{9}{4}\right) \cdot (x - 1, y - 1, z - 1) = 0$

Is the answer you have found a unique plane.

Yes.

C. Intersection of Two Planes

1

2.36: Intersection of Two Planes

The intersection of two distinct planes is always a line.

Example 2.37

The intersection of two planes is:

A.

2

Example 2.38

Find the equation of the line where the plane $2x - y - 4z = 8$ and $4x + y + 2z = 10$ intersect.

Substitute $z = 0$, and solve the resulting system of equations in 2 variables:

$$(3, -2, 0)$$

Substitute $z = 1$, and solve the resulting system of equations in 2 variables:

$$\left(\frac{10}{3}, -\frac{16}{3}, 1\right)$$

Direction Vector:

$$\left(3 - \frac{10}{3}, -2 + \frac{16}{3}, 0 - 1\right) = \left(-\frac{1}{3}, \frac{10}{3}, -1\right)$$

Substitute $\vec{a} = (3, -2, 0)$, $\vec{b} = \left(-\frac{1}{3}, \frac{10}{3}, -1\right)$ in the equation of a line:

$$\vec{r} = \vec{a} + \lambda \vec{b} = (3, -2, 0) + \lambda \left(-\frac{1}{3}, \frac{10}{3}, -1\right) = \left(3 - \frac{\lambda}{3}, -2 + \frac{10\lambda}{3}, -\lambda\right)$$

D. Intersection of Line and Plane

3

Example 2.39

15: Find the distance between the point $(-1, -5, -10)$ and the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-12}{2}$ and the plane $x - y + z = 5$. (CBSE 2015)

Example 2.40

Which point on the plane $2x + 3y + 6z = 98$ is closest to the origin?

The point on the plane closest to any point lies on the line perpendicular to that point. The vector of coefficients $(2, 3, 6)$ is perpendicular to the plane.

$\therefore$ Both the origin, and $(2, 3, 6)$ lie on the line perpendicular to the plane.

The vector equation of a line is

$$r(t) = P + tv, \text{ where } P = (0, 0, 0), v = (2, 3, 6)$$

$$r = (0, 0, 0) + t(2, 3, 6)$$

$$r = t(2, 3, 6)$$

$$xi + yj + zk = 2t\hat{i} + 3t\hat{j} + 6t\hat{k}$$

Equating coefficients gives us:

$$x = 2t, y = 3t, z = 6t$$

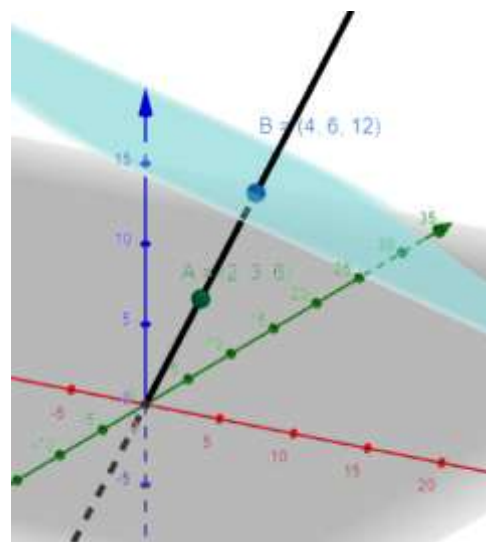
We find the intersection of the line and the plane. Substituting the above in the equation of the plane $2x + 3y + 6z = 98$:

$$2(2t) + 3(3t) + 6(6t) = 98 \Rightarrow 49t = 98 \Rightarrow t = 2$$

Hence,

$$x = 2t = 4, y = 3t = 6, z = 6t = 12 \Rightarrow (x, y, z) = (4, 6, 12)$$

And this is the point on the plane closest to the origin.



E. Distance of a Point from a Plane

(Calculator) Example 2.41

Find the shortest distance from $(-1.2, -12.5, 6.1)$ to the plane $2x + 5y - z = -8$.

The point on the plane closest to any point lies on the line perpendicular to that point.

The vector of coefficients $(2, 5, -1)$ is perpendicular to the plane.

∴ Both $(-1.2, -12.5, 6.1)$, and $(2, 5, -1)$ lie on the line perpendicular to the plane.

Substitute $P = (-1.2, -12.5, 6.1)$, $\mathbf{v} = (2, 5, -1)$ in the equation of a line:

$$\vec{r} = (-1.2, -12.5, 6.1) + t(2, 5, -1)$$

$$(x, y, z) = (-1.2 + 2t, -12.5 + 5t, 6.1 - t)$$

Find the intersection of the line and the plane. Substituting the above in the equation of the plane $2x + 3y + 6z = 98$:

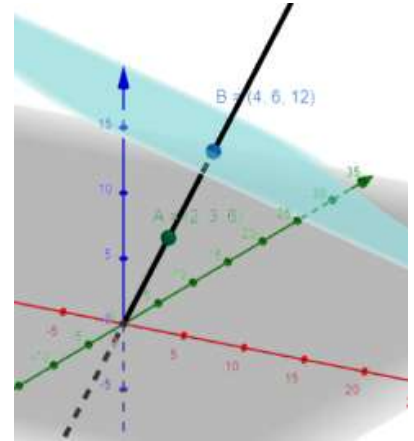
$$2(-1.2 + 2t) + 5(-12.5 + 5t) - 1(6.1 - t) = -8 \Rightarrow t = 2.1$$

Hence,

$$(x, y, z) = (-1.2 + 2t, -12.5 + 5t, 6.1 - t) = (3, -2, 4)$$

Use the distance formula to find the distance between $(-1.2, -12.5, 6.1)$ and $(3, -2, 4)$

$$d = \sqrt{(-1.2 - 3)^2 + (-12.5 + 2)^2 + (6.1 - 4)^2} = 11.502$$



2.42: Distance of a Point from a Plane

The distance of the point (x_1, y_1, z_1) from the plane $Ax + By + Cz = D$ is

$$\frac{|Ax_1 + By_1 + Cz_1 - D|}{\sqrt{A^2 + B^2 + C^2}}$$

The points (x_1, y_1, z_1) and (A, B, C) lie on a line perpendicular to the plane with equation:

$$(x, y, z) = (x_1, y_1, z_1) + \lambda(A, B, C) = (x_1 + \lambda A, y_1 + \lambda B, z_1 + \lambda C)$$

Substitute the above into $Ax + By + Cz = D$ to determine the point of intersection of the line and the plane:

$$A(x_1 + \lambda A) + B(y_1 + \lambda B) + C(z_1 + \lambda C) = D$$

Move the constants to the other side, and factor λ on the LHS:

$$\lambda(A^2 + B^2 + C^2) = D - Ax_1 - By_1 - Cz_1$$

Solve for λ :

$$\lambda = \frac{D - Ax_1 - By_1 - Cz_1}{A^2 + B^2 + C^2}$$

Equation I

We now want to find the distance between the points for the value of λ above.

$$(x_1, y_1, z_1) \text{ and } (x_1 + \lambda A, y_1 + \lambda B, z_1 + \lambda C)$$

By the distance formula, the distance is:

$$\sqrt{(x_1 - x_1 + \lambda A)^2 + (y_1 - y_1 + \lambda B)^2 + (z_1 - z_1 + \lambda C)^2}$$

This simplifies to:

$$\sqrt{(\lambda A)^2 + (\lambda B)^2 + (\lambda C)^2}$$

Which on expansion gives:

$$\sqrt{\lambda^2(A^2 + B^2 + C^2)} = |\lambda| \sqrt{A^2 + B^2 + C^2}$$

Substitute the value of λ from Equation I:

$$\left| \frac{D - Ax_1 - By_1 - Cz_1}{A^2 + B^2 + C^2} \right| \sqrt{A^2 + B^2 + C^2}$$

Which simplifies to:

$$\left| \frac{D - Ax_1 - By_1 - Cz_1}{\sqrt{A^2 + B^2 + C^2}} \right|$$

And using the property that $|x| = |-x|$, we get the expression

$$\left| \frac{Ax_1 + By_1 + Cz_1 - D}{\sqrt{A^2 + B^2 + C^2}} \right|$$

Which is what we wanted to show.

4

Example 2.43

The formula that we found above is for distance of a point from a plane. We had earlier found a formula for the distance of a plane from the origin. What is the connection between the two formulas. Justify.

Note that

Formula 1 =

Formula 2 =

Substitute the coordinates of the origin (0,0,0) in the formula for the distance of a point from a plane:

And we get the formula that we had earlier.

Hence, Formula 1 is a special case of Formula 2.

2.3 Normal Form

A. Normal Form

2.44: Revision: Direction Angles and Direction Cosines

If O is the origin, then the angles made by $\vec{OP}$ with the positive direction of the axes are the direction angles. The cosines of the direction angles are the direction cosines:

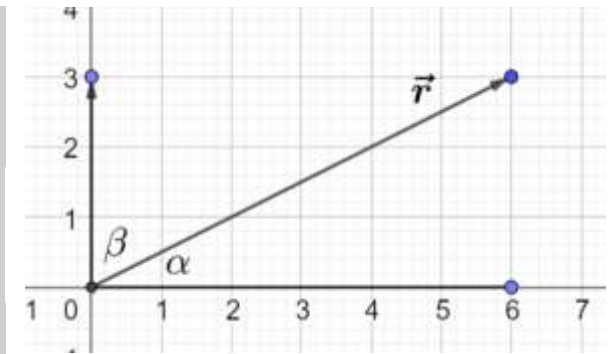
$$\text{Angle with } x\text{-axis} = l = \frac{x}{|\vec{r}|} = \cos \alpha$$

$$\text{Angle with } y\text{-axis} = m = \frac{y}{|\vec{r}|} = \cos \beta$$

$$\text{Angle with } z\text{-axis} = n = \frac{z}{|\vec{r}|} = \cos \gamma$$

Sum of squares of direction cosines is one:

$$l^2 + m^2 + n^2 = 1$$



Example 2.45

Convert the direction ratios (A, B, C) to direction cosines.

Divide by the magnitude of the direction ratio vector.

$$\underbrace{(A, B, C)}_{\text{Direction Ratios}} \rightarrow \frac{(A, B, C)}{\sqrt{A^2 + B^2 + C^2}} \quad \underbrace{\hspace{1cm}}_{\text{Direction Cosines}}$$

2.46: Normal Form: Cartesian Equation:

$$lx + my + nz = p$$

Begin with the Cartesian equation of a plane in general form:

$$Ax + By + Cz = D$$

Divide both sides by the magnitude of the direction ratio vector to convert the direction ratios to direction cosines:

$$\frac{A}{\sqrt{A^2 + B^2 + C^2}}x + \frac{B}{\sqrt{A^2 + B^2 + C^2}}y + \frac{C}{\sqrt{A^2 + B^2 + C^2}}z = \frac{D}{\sqrt{A^2 + B^2 + C^2}}$$

Since each the coefficients on the LHS are direction cosines, use the conventional notation (l, m, n) for them, and let $\frac{D}{\sqrt{A^2 + B^2 + C^2}} = p$:

$$lx + my + nz = p$$

Example 2.47

Write the plane $2x - 3y + z = 3$ in normal form.

Divide both sides by the magnitude of the direction ratio vector $= \sqrt{2^2 + (-3)^2 + 1^2} = \sqrt{14}$:

$$\frac{2}{\sqrt{14}}x - \frac{3}{\sqrt{14}}y + \frac{z}{\sqrt{14}} = \frac{3}{\sqrt{14}}$$

2.48: Distance of a Plane from the Origin

The distance of a plane in normal form $(lx + my + nz = p, p > 0)$ from the origin is

$$p$$

$$\begin{aligned} l &= \frac{x}{|\vec{r}|} = \cos \alpha \Rightarrow x = l|\vec{r}| \\ m &= \frac{y}{|\vec{r}|} = \cos \beta \Rightarrow y = m|\vec{r}| \\ n &= \frac{z}{|\vec{r}|} = \cos \gamma \Rightarrow z = n|\vec{r}| \end{aligned}$$

The equation of a plane in normal form is:

$$lx + my + nz = p$$

Substitute $x = l|\vec{r}|, y = m|\vec{r}|, z = n|\vec{r}|$ in the above:

$$l(l|\vec{r}|) + m(m|\vec{r}|) + n(n|\vec{r}|) = p$$

Multiply:

$$l^2|\vec{r}| + m^2|\vec{r}| + n^2|\vec{r}| = p$$

Factor $|\vec{r}|$:

$$|\vec{r}|(l^2 + m^2 + n^2) = p$$

Since l, m and n are the direction cosines and the sum of their squares is 1, substitute $l^2 + m^2 + n^2 = 1$

$$|\vec{r}| = p$$

And recall that $\vec{r}$ is the vector perpendicular to the plane with tail at the origin and origin and tip at the plane. Hence

$$\text{Length of } \vec{r} = |\vec{r}| = p = \text{Distance of Plane from origin}$$

Example 2.49

A. Determine the distance of the plane $x - 4y + 2z = 7$ from the origin.

- B. 12: Write the distance of the following plane from the origin: $2x - y + 2z + 1 = 0$. (CBSE 2010)
C. Find the length of the perpendicular drawn from the origin to the plane $2x - 3y + 6z + 21 = 0$ from the origin. (CBSE 2011)

Part A

Divide both sides by the magnitude of the direction ratio vector $= \sqrt{1^2 + (-4)^2 + 2^2} = \sqrt{21}$:

$$\frac{x - 4y + 2z}{\sqrt{21}} = \frac{7}{\sqrt{21}} \Rightarrow \text{Distance} = \frac{7}{\sqrt{21}}$$

Part B

$$2x - y + 2z = -1$$

Divide both sides by the magnitude of the direction ratio vector $= \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3$:

$$\frac{2x - y + 2z}{3} = -\frac{1}{3}$$

Distance is

$$\left| -\frac{1}{3} \right| = \frac{1}{3}$$

Part C

The length of the perpendicular to the plane from the origin is just the distance of the plane from the origin.

$$-2x + 3y - 6z = 21$$

Divide both sides by the magnitude of the direction ratio vector $= \sqrt{(-2)^2 + 3^2 + (-6)^2} = \sqrt{49} = 7$:

$$\frac{-2x + 3y - 6z}{7} = 3$$

Distance from the origin

$$= 3$$

2.50: Distance of a Plane from the Origin (General Cartesian Form)

The distance of the plane $Ax + By + Cz = D$ from the origin is

$$\frac{D}{\sqrt{A^2 + B^2 + C^2}}$$

Begin with the Cartesian equation of a plane in general form:

$$Ax + By + Cz = D$$

Divide both sides by the magnitude of the direction ratio vector to convert the direction ratios to direction cosines:

$$\frac{A}{\sqrt{A^2 + B^2 + C^2}}x + \frac{B}{\sqrt{A^2 + B^2 + C^2}}y + \frac{C}{\sqrt{A^2 + B^2 + C^2}}z = \frac{D}{\sqrt{A^2 + B^2 + C^2}}$$

Since each the coefficients on the LHS are direction cosines, use the conventional notation (l, m, n) for them, d let $\frac{D}{\sqrt{A^2 + B^2 + C^2}} = p$:

$$lx + my + nz = \frac{D}{\sqrt{A^2 + B^2 + C^2}}$$

Since the above is in normal, the distance from the origin is:

$$\frac{D}{\sqrt{A^2 + B^2 + C^2}}$$

2.51: Normal Form: Vector Equation:

$$\hat{n} \cdot \vec{r} = p$$

Begin with the Cartesian equation of a plane in normal form:

$$lx + my + nz = p$$

Factor and rewrite the LHS as the dot product of two vectors:

$$(l, m, n) \cdot (x, y, z) = p$$

Since l, m, n are direction cosines $l^2 + m^2 + n^2 = 1$, and hence (l, m, n) is a unit vector.

Let $\vec{r} = (x, y, z)$.

Then, the above equation becomes:

$$\hat{n} \cdot \vec{r} = p$$

Concept Check 2.52

- What is the difference between the general form, and the normal form of the equation of a plane?
- How do you convert the equation of a plane from general form to normal form?

Part A

- The LHS of the normal form has a unit vector multiplying the vector (x, y, z) , whereas the general form has some non-zero multiple of it.
- The RHS of the normal form indicates the distance from the origin, but this may not be true in the general form.

Part B

Divide both sides by the magnitude of the direction ratios.

Example 2.53

- Find the vector equation of a plane which is a distance of 5 units from the origin, and its normal vector is $(2, -3, 6)$. (CBSE 2016)
- Write the vector equation of a plane which is at a distance of $5\sqrt{3}$ from the origin and the normal to which is equally inclined to the coordinate axes. (CBSE 2016)

Part A

$$\begin{aligned}\hat{n} \cdot \vec{r} &= p \\ \frac{\vec{n}}{|\vec{n}|} \cdot \vec{r} &= p \\ \frac{(2, -3, 6)}{\sqrt{2^2 + (-3)^2 + 6^2}} \cdot \vec{r} &= 5 \\ (2, -3, 6) \cdot \vec{r} &= 5\sqrt{49} \\ (2, -3, 6) \cdot \vec{r} &= 35\end{aligned}$$

Part B

The direction cosines meet the constraint:

$$l^2 + m^2 + n^2 = 1$$

Since they are equal:

$$3l^2 = 1 \Rightarrow l^2 = \frac{1}{3} \Rightarrow l = \frac{1}{\sqrt{3}}$$

Substitute $l = m = n = \frac{1}{\sqrt{3}}$ in $\hat{n} \cdot \vec{r} = p$:

$$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \cdot \vec{r} = 5\sqrt{3}$$

Multiply by $\sqrt{3}$ on both sides:

$$(1, 1, 1) \cdot \vec{r} = 15$$

2.54: Distance between Two Parallel Planes: Normal Form

Given two parallel planes in normal form $P_1: lm + my + nz = p_1$, and $P_2: lm + my + nz = p_2$, the distance between them is:

$$|p_1 - p_2|$$

Now the distance of P_1 from the origin is $|p_1|$. And the distance of P_2 from the origin is $|p_2|$.

Hence, the origin, the closest point of P_1 from the origin, and the closest point of P_2 from the origin lie on a straight line.

And, hence, the distance between the two planes is:

$$|p_1 - p_2|$$

Concept Check 2.55

Does the formula $|p_1 - p_2|$ cover the case where the two planes lie on opposite sides of the origin.

Yes, it does cover that case.

$$\begin{aligned}lm + my + nz &= 3 \\lm + my + nz &= -5\end{aligned}$$

$$|3 - (-5)| = 8$$

Example 2.56

Find the distance between the planes given below:

$$\begin{aligned}\frac{1}{5}x - \frac{2}{\sqrt{5}}y + \frac{2}{5}z &= 7 \\ \frac{1}{5}x - \frac{2}{\sqrt{5}}y + \frac{2}{5}z &= -12\end{aligned}$$

Check the sum of the direction ratios:

$$\left(\frac{1}{5}\right)^2 + \left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{2}{5}\right)^2 = \frac{1}{25} + \frac{4}{5} + \frac{4}{25} = \frac{4}{5} + \frac{5}{25} = \frac{4}{5} + \frac{1}{5} = 1$$

Since the sum is 1, the equations are already in normal form, and hence the distance between the planes is:

$$|p_1 - p_2| = |7 - (-12)| = 19$$

Example 2.57

$$Ax + By + Cz = D_1$$

$$\frac{Ax + By + Cz}{\sqrt{A^2 + B^2 + C^2}} = \frac{D}{\sqrt{A^2 + B^2 + C^2}}$$

$$\frac{qAx + qBy + qCz}{\sqrt{(qA)^2 + (qB)^2 + (qC)^2}} = \frac{qD}{\sqrt{q^2A^2 + q^2B^2 + q^2C^2}} = \frac{qD}{\sqrt{q^2(A^2 + B^2 + C^2)}} = |q| \frac{D}{\sqrt{A^2 + B^2 + C^2}}$$

$$\frac{qAx + qBy + qCz}{|q|\sqrt{A^2 + B^2 + C^2}} = \frac{D_2}{|q|\sqrt{A^2 + B^2 + C^2}}$$

Hence, if the direction ratios are scaled by a factor of q , the magnitude of the normal vector is also scaled by q .

Example 2.58

The two planes $\frac{1}{3}x - 4y + 7z = p_1$ and $\frac{3}{5}x - by + cz = p_2$ are parallel. Let the distance between the planes be D . Find $D + b + c$.

The magnitude of the direction ratio vector:

$$\sqrt{\left(\frac{1}{3}\right)^2 + (-4)^2 + 7^2} = \sqrt{\frac{1}{9} + 16 + 49} = \sqrt{\frac{586}{9}} = \frac{\sqrt{586}}{3}$$

Instead of dividing both sides by the magnitude, multiply both sides by the reciprocal of the number above:

$$\frac{\left(\frac{3}{\sqrt{586}}\right)\left(\frac{1}{3}x - 4y + 7z\right)}{\frac{\sqrt{586}}{3}} = \frac{3}{\sqrt{586}} \times p_1$$

$$\frac{1}{3}a = \frac{3}{5} \Rightarrow a = \frac{9}{5}$$

$$\frac{9}{5}\left(\frac{1}{3}, -4, 7\right) = \left(\frac{3}{5}, -\frac{36}{5}, \frac{63}{5}\right) = \left(\frac{3}{5}, b, c\right)$$

$$b = -\frac{36}{5}, c = \frac{63}{5}$$

$$\frac{3}{5}x + \frac{36}{5}y + \frac{63}{5}z = p_2$$

Divide both sides by the magnitude of the direction vector $\frac{\sqrt{586}}{3} \times \frac{9}{5} = \frac{3\sqrt{586}}{5}$:

$$\left(\frac{5}{3\sqrt{586}}\right)\left(\frac{3}{5}x + \frac{36}{5}y + \frac{63}{5}z\right) = \frac{5}{3\sqrt{586}} \times p_2$$

The distance between the planes is:

$$\left| \frac{3p_1}{\sqrt{586}} - \frac{5p_2}{3\sqrt{586}} \right|$$

Finally, the quantity that we want is:

$$D + b + c = \left| \frac{3p_1}{\sqrt{586}} - \frac{5p_2}{3\sqrt{586}} \right| - \frac{36}{5} + \frac{63}{5} = \left| \frac{3p_1}{\sqrt{586}} - \frac{5p_2}{3\sqrt{586}} \right| + \frac{27}{5}$$

5

Example 2.59

1: Find the distance between the planes $2x - y + 2z = 5$ and $5x - 2.5y + 5z = 20$. (CBSE 2017)

Example 2.60

Find the distance between two parallel planes in general form. Assume variables as necessary.

Two planes are parallel when their normal vectors are oriented in the same direction. Hence, we can write the planes in general form as:

$$\begin{aligned} Ax + By + Cz &= D_1 \\ Ax + By + Cz &= D_2 \end{aligned}$$

Converting the above from general form to normal form:

$$\begin{aligned} \frac{Ax + By + Cz}{\sqrt{A^2 + B^2 + C^2}} &= \frac{D_1}{\sqrt{A^2 + B^2 + C^2}} \\ \frac{Ax + By + Cz}{\sqrt{A^2 + B^2 + C^2}} &= \frac{D_2}{\sqrt{A^2 + B^2 + C^2}} \end{aligned}$$

Since the above are now in normal form, the distance between them is:

$$\left| \frac{D_1}{\sqrt{A^2 + B^2 + C^2}} - \frac{D_2}{\sqrt{A^2 + B^2 + C^2}} \right|$$

Which simplifies to:

$$= \left| \frac{D_1 - D_2}{\sqrt{A^2 + B^2 + C^2}} \right|$$

Do you need to remember the “formula” above?

No. It is better to convert to normal form, and use the simpler formula.

2.4 Intercept Form

A. Intercept Form

2.61: Intercept Form: Cartesian Equation

If (a, b, c) are the x , y , and z intercepts respectively of a plane, then it has equation:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

Begin with the Cartesian equation of a plane in general form:

$$\underbrace{Ax + By + Cz = D}_{\text{Equation 1}}$$

At the x – axis, we must have $y = 0, z = 0$, which when we substitute in the above equation:

$$Ax + 0y + 0z = D \Rightarrow Ax = D \Rightarrow x = \frac{D}{A}$$

Similarly:

$$\begin{aligned} y - \text{intercept: } 0x + By + 0z &= D \Rightarrow By = D \Rightarrow y = \frac{D}{B} \\ z - \text{intercept: } 0x + 0y + Cz &= D \Rightarrow Cz = D \Rightarrow z = \frac{D}{C} \end{aligned}$$

Hence, we say that the intercepts are:

$$x = \frac{D}{A}, y = \frac{D}{B}, z = \frac{D}{C}$$

Now go back to Equation I ($Ax + By + Cz = D$) and divide both sides by D:

$$\frac{Ax}{D} + \frac{By}{D} + \frac{Cz}{D} = 1 \Rightarrow \frac{x}{\frac{D}{A}} + \frac{y}{\frac{D}{B}} + \frac{z}{\frac{D}{C}} = 1$$

Use a change of variable. Substitute $\frac{D}{A} = a, \frac{D}{B} = b, \frac{D}{C} = c$:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

Which is the intercept form of the equation of the plane with (a, b, c) as the x, y , and z intercepts respectively.

Example 2.62

What is the sum of the intercepts that the plane $\frac{3}{5}x + \frac{2}{3}y - \frac{5}{9}z = 11$ makes with the axes?

$$\begin{aligned} \frac{3}{55}x + \frac{2}{33}y - \frac{5}{99}z &= 1 \\ \frac{x}{\frac{55}{3}} + \frac{y}{\frac{33}{2}} - \frac{z}{\frac{99}{5}} &= 1 \end{aligned}$$

Sum of the intercepts is:

$$\frac{55}{3} + \frac{33}{2} - \frac{99}{5} = \frac{550}{30} + \frac{495}{30} - \frac{594}{30} = \frac{451}{30}$$

Example 2.63

Find the sum of the intercepts cut off by the plane $2x + y - z = 5$ on the coordinate axes. (CBSE 2015, 2011)

Divide both sides by 5:

$$\frac{2x}{5} + \frac{y}{5} - \frac{z}{5} = 1$$

Write the equation in intercept form:

$$\frac{x}{\frac{5}{2}} + \frac{y}{5} - \frac{z}{5} = 1$$

Sum of intercepts

$$= \frac{5}{2} + 5 - 5 = \frac{5}{2}$$

2.64: Intercept Form: Vector Equation

$$\vec{n} \cdot \vec{r} = 1$$

Begin with the Cartesian equation of a plane in intercept form:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

Factor:

$$\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right) \cdot (x, y, z) = 1$$

Substitute $\vec{n} = \left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right), \vec{r} = (x, y, z)$

$$\vec{n} \cdot \vec{r} = 1$$

Example 2.65

- A. Find the vector equation of the plane with intercepts 3, -4 and 2 on X, Y and Z axes. (CBSE 2016)
B. Write the sum of intercepts cut off by the plane $\vec{r} \cdot (2, 1, -1) - 5 = 0$ on the three axes. (CBSE 2015)
C. Find the intercepts cut off by the plane $\vec{r} \cdot \left(\frac{2}{3}, -\frac{3}{5}, \frac{5}{7}\right) - \frac{1}{3} = 0$ on the three axes.

Part A

$$\left(\frac{1}{3}, \frac{1}{-4}, \frac{1}{2}\right) \cdot (x, y, z) = 1$$

Substitute $\vec{r} = (x, y, z)$

$$(4, -3, 6) \cdot \vec{r} = 12$$

Part B

$$\vec{r} \cdot (2, 1, -1) = 5$$

$$\vec{r} \cdot \left(\frac{2}{5}, \frac{1}{5}, -\frac{1}{5}\right) = 1$$

The intercepts are then the reciprocals of the direction ratios:

$$x = \frac{5}{2}, y = 5, z = -5$$

The sum is:

$$\frac{5}{2} + 5 - 5 = \frac{5}{2}$$

Part C

$$\vec{r} \cdot \left(\frac{2}{3}, -\frac{3}{5}, \frac{5}{7}\right) = \frac{1}{3}$$

Multiply by 3 both sides:

$$\vec{r} \cdot \left(2, -\frac{9}{5}, \frac{15}{7}\right) = 1$$

The intercepts are then the reciprocals of the direction ratios:

$$x = \frac{1}{2}, y = -\frac{5}{9}, z = \frac{7}{15}$$

B. Revision

2.66: Plane Equations: Summary

- $\vec{r} = P_0(x_0, y_0, z_0)$ is a point on the plane
- $P(x, y, z)$ is a general point on the plane
- $\vec{n} = (A, B, C)$ is a vector normal to the plane
- $\hat{n}$ is a unit vector in the direction of $\vec{n}$
- (A, B, C) are the direction ratios of the normal to the plane
- (l, m, n) are the direction cosines of the normal to the plane
- (a, b, c) are the intercepts of the plane on the x, y, z axes respectively

Form	Vector Equation	Cartesian Equation
Point	$\vec{n} \cdot \overrightarrow{P_0P} = 0$	$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$
General	$\vec{n} \cdot \vec{r} = D$	$Ax + By + Cz = D$
Normal	$\hat{n} \cdot \vec{r} = p$	$lx + my + nz = p$
Intercept Form	$\vec{n} \cdot \vec{r} = 1$	$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Concept Check 2.67

How do you convert the Cartesian Equation:

- A. general form to normal form
- B. general form to intercept form
- C. Point form to general form
- D. Point form to normal form
- E. Point form to intercept form
- F. Normal form to general form
- G. Normal form to intercept form

H. Intercept Form to General Form

Part A

Divide both sides of $Ax + By + Cz = D$ by the magnitude of the normal vector $= \sqrt{A^2 + B^2 + C^2}$ to get:

$$\frac{A}{\sqrt{A^2 + B^2 + C^2}}x + \frac{B}{\sqrt{A^2 + B^2 + C^2}}y + \frac{C}{\sqrt{A^2 + B^2 + C^2}}z = \frac{D}{\sqrt{A^2 + B^2 + C^2}}$$

Part B

Divide both sides of $Ax + By + Cz = D$ by D to get:

$$\frac{A}{D}x + \frac{B}{D}y + \frac{C}{D}z = 1 \Rightarrow \frac{x}{\frac{D}{A}} + \frac{y}{\frac{D}{B}} + \frac{z}{\frac{D}{C}} = 1$$

Part C

Expand the LHS, simplify and send the constant to the RHS

Part D

Convert to general form, and then convert from general form to point form.

Part E

Convert to general form, and then convert from general form to point intercept.

Part F

Multiply on both sides by any non-zero quantity

Part G

Divide both sides of $lx + my + nz = p$ by p to get $\frac{l}{p}x + \frac{m}{p}y + \frac{n}{p}z = 1$

Part H

Multiply sides of $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ by abc to get:

$$bcx + acy + abz = 1$$
$$A = bc, B = ac, C = ab$$

Example 2.68

Plane P is parallel to $-4x + 5y - 2z = 37$ and passes through $(-2, 8, 5)$. Write the Vector and Cartesian equations of P in point form, general form, normal form and intercept form.

Point Form

The Vector equation of a plane in point form is:

$$\vec{n} \cdot \overrightarrow{P_0P} = 0$$

Equation I

Note that $\vec{n} = (-4, 5, -2)$ and use $P_0(x_0, y_0, z_0) = (-2, 8, 5)$ and $P(x, y, z)$ to find:

$$\overrightarrow{P_0P} = (x + 2, y - 8, z - 5)$$

Equation III

Substitute Equation II and III in Equation I to get the Vector equation in point form:

$$(-4, 5, -2) \cdot (x + 2, y - 8, z - 5) = 0$$

To convert to the Cartesian form, expand the LHS using the dot product definition:

$$-4(x + 2) + 5(y - 8) - 2(z - 5) = 0$$

General Form

To convert to general form, use the distributive property on the LHS:

$$(-4x - 8) + (5y - 40) + (-2z + 10) = 0$$

Simplify to get the Cartesian equation in general form:

$$-4x + 5y - 2z = 38$$

Factor to get the vector version:

$$(-4, 5, -2) \cdot (x, y, z) = 38$$

Normal Form

The magnitude of the normal vector is:

$$\sqrt{(-4)^2 + 5^2 + (-2)^2} = \sqrt{45} = 3\sqrt{5}$$

To convert to normal form, divide both sides of the Cartesian equation from above by the magnitude of the normal vector:

$$-\frac{4}{3\sqrt{5}}x + \frac{5}{3\sqrt{5}}y - \frac{2}{3\sqrt{5}}z = \frac{38}{3\sqrt{5}}$$

Factor to get the vector form:

$$\left(-\frac{4}{3\sqrt{5}}, \frac{5}{3\sqrt{5}}, -\frac{2}{3\sqrt{5}}\right) \cdot (x, y, z) = \frac{38}{3\sqrt{5}}$$

Intercept Form

To convert to intercept form, divide both sides of the Cartesian equation in general form by 38:

$$-\frac{4}{38}x + \frac{5}{38}y - \frac{2}{38}z = 1$$

Rearrange in intercept form:

$$\frac{x}{-\frac{19}{2}} + \frac{y}{\frac{38}{5}} + \frac{z}{-\frac{19}{2}} = 1$$

Factor to get the vector version:

$$\left(\frac{1}{-\frac{19}{2}}, \frac{1}{\frac{38}{5}}, \frac{1}{-\frac{19}{2}}\right) \cdot (x, y, z) = 1$$

Example 2.69

The plane P passes through the point (a, b, c) and is parallel to the plane $\vec{r} \cdot (1, 1, 1) = 2$. Write the Vector and Cartesian equations of P in point form, general form, normal form and intercept form. (CBSE 2014, Adapted)

We have that plane $\vec{r} \cdot (1, 1, 1) = 2$ is parallel to plane P . Since the two planes are parallel, the direction ratios of their normal vectors are equivalent. Hence, the direction ratios of the normal vector to plane P are:

$$\vec{n} = (1, 1, 1)$$

Point Form

The Vector equation of a plane in point form is:

$$\vec{n} \cdot \overrightarrow{P_0P} = 0$$

Equation I

Note that $\vec{n} = (1, 1, 1)$ and use $P_0(x_0, y_0, z_0) = (a, b, c)$ and $P(x, y, z)$ to find:

Equation II

$$\overrightarrow{P_0P} = (x - a, y - b, z - c)$$

Equation III

Substitute Equation II and III in Equation I to get the Vector equation in point form:

$$(1,1,1) \cdot (x - a, y - b, z - c) = 0$$

To convert to Cartesian form, expand the LHS using the dot product definition:

$$(x - a) + (y - b) + (z - c) = 0$$

General Form

To convert to general form, move the constants to the RHS:

$$x + y + z = a + b + c$$

Factor to get the vector version:

$$(1,1,1) \cdot (x, y, z) = a + b + c$$

Normal Form

The magnitude of the normal vector is:

$$\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

To convert to normal form, divide both sides of the Cartesian equation from above by the magnitude of the normal vector:

$$\frac{x}{\sqrt{3}} + \frac{y}{\sqrt{3}} + \frac{z}{\sqrt{3}} = \frac{a + b + c}{\sqrt{3}}$$

Factor to get the vector form:

$$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \cdot (x, y, z) = \frac{a + b + c}{\sqrt{3}}$$

Intercept Form

To convert to intercept form, divide both sides of the Cartesian equation in general form by 38:

$$\frac{x}{\frac{a + b + c}{\sqrt{3}}} + \frac{y}{\frac{a + b + c}{\sqrt{3}}} + \frac{z}{\frac{a + b + c}{\sqrt{3}}} = 1$$

Factor to get the vector version:

$$\left(\frac{1}{a + b + c}, \frac{1}{a + b + c}, \frac{1}{a + b + c}\right) \cdot (x, y, z) = 1$$

2.5 Planes, Lines, Points

A. Angle between Two Planes

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2.70: Angle between Two Planes

8

Example 2.71

B. Angle between a Line and a Plane

9

2.72: Angle between a Line and a Planes

Example 2.73

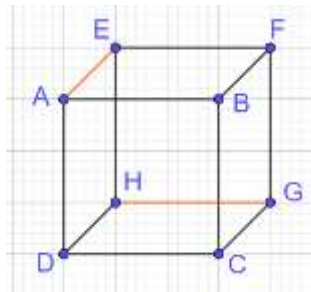
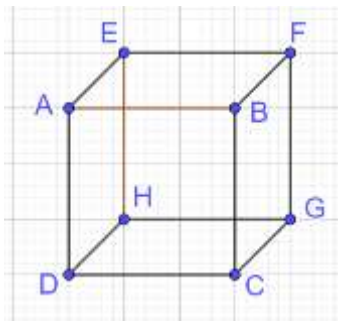
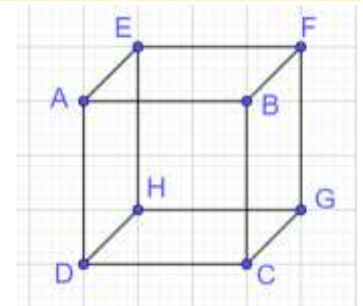
C. Coplanarity of Two Lines

Example 2.74

Coplanar lines are lines that are in the same plane. If two lines are not coplanar, what kind of lines are they? Give one example of such kind of lines.

Lines which are not coplanar are skew.

Example 2.75



$(AB, EH), (AB, FG), (AB, DH), (AB, CG)$

Similarly, every edge forms 4 pairs.

The total number of pairs formed is:

$$\frac{12 \times 4}{2} = 24$$

$(AE, HG), (AE, DC),$
 $(BF, AD), (BF, EH)$
 $(EF, AD), (EF, BC)$

2.76: Coplanarity of Two Lines

Two lines $\vec{r} = \vec{a}_1 + \lambda_1 \vec{a}$ and $\vec{r} = \vec{a}_2 + \lambda_2 \vec{b}$ are coplanar if:

$$(\vec{a}_1 - \vec{a}_2) \cdot (\vec{a} \times \vec{b}) = 0$$

$\vec{a}_1$ and $\vec{a}_2$ are points on the first and second line respectively.

If the two lines are coplanar, then they lie in the same plane, and so does the vector joining them:

$$\vec{a}_1 - \vec{a}_2$$

Again, if the lines are coplanar, $\vec{b}_1$ and $\vec{b}_2$ are vectors in the plane. The cross product $\vec{b}_1 \times \vec{b}_2$ must be perpendicular to the plane.

And finally, the dot product of a vector perpendicular to the plane with a vector in the plane must be zero:

$$\underbrace{(\vec{a}_1 - \vec{a}_2)}_{\substack{\text{Vector} \\ \text{in the Plane}}} \cdot \underbrace{(\vec{b}_1 \times \vec{b}_2)}_{\substack{\text{Vector} \perp \\ \text{to the plane}}} = 0$$

$$\vec{c} \cdot (\vec{a} \times \vec{b}) = \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Example 2.77

Show that the lines $\frac{5-x}{-4} = \frac{y-7}{4} = \frac{z+3}{-5}$ and $\frac{x-8}{7} = \frac{2y-8}{2} = \frac{z-5}{3}$ are coplanar.

$$\begin{aligned} \frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5} &\Rightarrow \vec{r} = (5, 7, -3) + \lambda_1(4, 4, -5) \\ \frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3} &\Rightarrow \vec{r} = (8, 4, 5) + \lambda_2(7, 1, 3) \end{aligned}$$

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Example 2.78

$$\begin{aligned} \underbrace{ax_1 + by_1 + cz_1}_{\text{Equation I}} &= p \\ \underbrace{ax_2 + by_2 + cz_2}_{\text{Equation II}} &= p \end{aligned}$$

Subtract Equation II from Equation I:

$$\begin{aligned} a(x_1 - x_2) + b(y_1 - y_2) + c(z_1 - z_2) &= 0 \\ [a, b, c] \cdot [x_1 - x_2, y_1 - y_2, z_1 - z_2] &= 0 \end{aligned}$$

2.6 Further Topics

2.79: Distance between two skew lines

I: Let plane A be the unique plane containing line 1 and parallel to line 2.

Find the direction vectors of line 1 and line 2. (Line 1 lies in Plane A, hence the direction vector of line 1 also lies in plane A.

Line 2 is parallel to plane A. Hence, the direction vector of line 2 lies in plane A)

Find the normal vector $\vec{n}$ of plane A by taking the cross product of the above direction vectors.

II: Find the distance between a point on Line 2, and plane A

Let P_1 be a point on line 1, and P_2 be a point on Line 2

Find the vector $\overrightarrow{P_1P_2}$

Find the component of $\overrightarrow{P_1P_2}$ in the direction of $\vec{n}$ by using the formula

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Aziz Manva (azizmanva@gmail.com)

$$\overrightarrow{P_1P_2} \cdot \frac{\vec{n}}{|\vec{n}|}$$

80 Examples